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Brand Names

Ativan, Loreev XR

Indication Specific Dosing

General dosing and discontinuation recommendations
The risks and benefits of ongoing benzodiazepine use should be assessed every 3 months or as clinically indicated.
Those who are candidates for discontinuation should be assessed for risks of withdrawal prior to stopping the medication; individuals who are likely to be physically dependent on the benzodiazepine should be tapered off the medication rather than having an abrupt discontinuation. While most tapers can be managed outpatient, patients at imminent risk of harm related to continued use (e.g., drug interactions, falls, or overdose), those with concomitant physical or mental health conditions that may complicate the safe tapering, and those with current or anticipated severe or complicated withdrawals may require inpatient care.
Tapering strategies should be patient-specific and may be adjusted as needed to address withdrawal symptoms. Initial dose reductions of 5% are recommended, with an assessment of tolerability made. In general, the pace of benzodiazepine tapering should include dose reductions of 5% to 10% every 2 to 4 weeks and should not exceed 25% every 2 weeks.
Individuals at risk of physical dependence, such as those taking a higher dose for more than a year, may require a conservative taper of 5% to 10% every 6 to 8 weeks or slower, depending on symptoms. Those with lower risks of physical dependence or who tolerate the initial dose reduction may be able to reduce the dose by 10% to 25% every 2 to 4 weeks.
If withdrawal symptoms occur and interfere with the taper (e.g., sleep difficulty or anxiety), the taper may be paused or further slowed. Adjunctive medications may also be considered.

For the treatment of generalized anxiety disorder (GAD) or for the short-term relief of the symptoms of anxiety

Oral dosage (immediate-release)

Adults

2 to 3 mg/day PO in 2 to 3 divided doses, initially. May increase the dose gradually as needed. Use the lowest effective dose. When a higher dosage is needed, increase the evening dose before the daytime doses. Usual dose range: 2 to 6 mg/day. Max: 10 mg/day. To discontinue, taper the dose gradually.

Geriatric Adults

1 to 2 mg/day PO in 2 to 3 divided doses, initially. May increase the dose gradually as needed. Geriatric adults are more sensitive to the effects of benzodiazepines. Use the lowest effective dose. When a higher dosage is needed, increase the evening dose before the daytime doses. Usual adult dose range: 2 to 6 mg/day. Max: 10 mg/day. To discontinue, taper the dose gradually.

Children and Adolescents 12 to 17 years

2 to 3 mg/day PO in 2 to 3 divided doses, initially. May increase the dose gradually as needed. Use the lowest effective dose. When a higher dosage is needed, increase the evening dose before the daytime doses. Usual dose range: 2 to 6 mg/day. Max: 10 mg/day. To discontinue, taper the dose gradually.

Children† 1 to 11 years

Dosage not available for anxiety disorders; however, lorazepam 0.025 to 0.05 mg/kg/dose PO as needed (no more frequently than every 4 hours) has been used in burn patients with anxiety related to being in the hospital, dressing changes, etc. In older pediatric patients, the daily dosage for anxiety disorders is typically divided into 2 to 3 doses with a maximum of 10 mg/day.

Oral dosage [extended-release, (i.e., Loreev XR)] in persons receiving stable, evenly divided, 3 times daily dosing with immediate-release products

Adults

Initiate ER dosing by calculating the total daily dose of lorazepam IR (immediate-release), then give PO once daily in the morning, starting the day after IR product discontinuation. Usual range: 2 to 6 mg PO once daily. Use the lowest effective dose. If an increase is needed, discontinue the ER capsules and increase the dosage using lorazepam IR. Once adequate response is achieved, resume

treatment with the ER capsules. Max: 10 mg/day. To discontinue, taper or decrease dose gradually.

Geriatric Adults

Initiate ER dosing by calculating the total daily dose of lorazepam IR (immediate-release), then give PO once daily in the morning, starting the day after IR product discontinuation. Usual range: 2 to 6 mg PO once daily. Use the lowest effective dose. If an increase is needed, discontinue the ER capsules and increase the dosage using lorazepam IR. Once adequate response is achieved, resume treatment with the ER capsules. Max: 10 mg/day. To discontinue, taper or decrease dose gradually.

For the short-term treatment of insomnia due to anxiety or transient situational stress

Oral dosage

Adults

2 to 4 mg PO at bedtime as needed. Debilitated adults may respond sufficiently to a lower dosage.

Geriatric Adults

1 to 2 mg PO at bedtime as needed, initially. Titrate slowly if clinically indicated. Geriatric adults are more sensitive to the effects of benzodiazepines. Use the lowest effective dose.

Children and Adolescents 12 to 17 years

2 to 4 mg PO at bedtime as needed. Debilitated patients may respond sufficiently to a lower dosage.

For procedural sedation or preoperative sedation induction

For operative amnesia induction in adult patients

Intravenous dosage

Adults

Up to 0.05 mg/kg IV (Max: 4 mg) during surgery or the procedure.

For preoperative sedation induction and/or relief of preoperative anxiety in adult patients

Intravenous or Intramuscular dosage

Adults

0.044 mg/kg IV (Max: 2 mg) 15 to 20 minutes prior to surgery or the procedure. Alternatively, 0.05 mg/kg IM (Max: 4 mg) administered 2 hours prior to surgery or the procedure.

For amnesia induction† and anxiety relief in pediatric patients

Oral dosage

Infants, Children, and Adolescents

0.05 mg/kg PO as a single dose (Max: 4 mg) 45 to 90 minutes prior to procedure. Dose range: 0.02 to 0.09 mg/kg/dose.

Intravenous or Intramuscular dosage

Infants, Children, and Adolescents

0.05 to 0.1 mg/kg IV or IM as a single dose (Max: 2 to 4 mg). Dose range: 0.02 to 0.1 mg/kg/dose. For optimum lack of recall, administer IV dose 15 to 20 minutes prior to procedure and IM dose 2 hours prior to procedure.

For sedation maintenance† in mechanically-ventilated patients

Intermittent Intravenous dosage

Adults

0.044 mg/kg/dose (e.g., 2 to 4 mg) IV every 2 to 4 hours, as needed; however, the required dosage is highly variable and should be titrated to desired degree of sedation. A single dose should not exceed 4 mg IV.

Infants, Children, and Adolescents

0.05 mg/kg/dose IV every 2 to 8 hours as needed (Max initial dose: 2 mg). Dose range: 0.025 to 0.1 mg/kg/dose. Due to a prolonged half-life, infants may require doses at less frequent intervals (e.g., every 6 to 8 hours) compared to children and adolescents.

Neonates

0.05 mg/kg/dose IV every 2 to 8 hours as needed. Dose range: 0.025 to 0.1 mg/kg/dose. Due to a prolonged half-life, neonates may require doses at less frequent intervals (e.g., every 6 to 8 hours) compared to children and adolescents.

Continuous IV Infusion† dosage

Adults

The usual dosage range is 0.5 to 8 mg/hour (or 0.01 to 0.1 mg/kg/hour); titrated to effect. The required dosage is highly variable and should be titrated to desired degree of sedation. A loading dose (i.e., 2 to 4 mg IV) is generally required.

Infants, Children, and Adolescents

Limited published data are available in the pediatric population. An initial infusion rate of 0.025 to 0.05 mg/kg/hour IV is recommended by some experts. Max initial rate: 2 mg/hour. A published sedation protocol for pediatric mechanically ventilated patients recommends an initial infusion rate of 0.01 mg/kg/hour IV. If 3 intermittent boluses of lorazepam are needed in a 6 hour time period, increase the infusion rate by 0.005 mg/kg/hour (50% of initial rate). If no additional boluses are needed, consider reducing the infusion rate. Titrate to desired level of sedation. High doses and prolonged infusions may increase the risk of propylene glycol toxicity; monitor patients carefully.

For the treatment of status epilepticus

Intravenous or Intramuscular dosage

Adults

0.1 mg/kg/dose (Max: 4 mg/dose) IV or IM as a single dose. May repeat dose once in 5 to 15 minutes.

Infants†, Children†, and Adolescents†

0.05 to 0.1 mg/kg/dose (Max: 4 mg/dose) IV or IM as a single dose. May repeat dose once in 5 to 15 minutes.

Neonates†

0.05 to 0.1 mg/kg/dose IV or IM as a single dose. May repeat dose once in 10 to 15 minutes.

Intranasal dosage†

Infants, Children, and Adolescents 2 months to 14 years

0.1 mg/kg/dose (Max: 4 mg/dose) intranasally in 1 nostril as a single dose. May repeat dose after 10 minutes if seizure continues. If using an intranasal benzodiazepine, guidelines recommend intranasal midazolam given better evidence supporting use.

For the treatment of alcohol withdrawal†

For the treatment of non-severe alcohol withdrawal†

Oral dosage (immediate-release, fixed-dose)

Adults

2 mg PO every 8 hours on days 1 and 2, then 1 mg PO every 8 hours on day 3, then 1 mg PO every 12 hours on day 4, and then 1 mg PO once daily at bedtime on day 5.

Oral dosage (immediate-release, symptom-triggered)

Adults

2 mg PO every 6 hours as needed on days 1 and 2, then 1 mg PO every 8 hours as needed on day 3, and then 1 mg PO every 12 hours as needed on days 4 and 5.

For the treatment of severe alcohol withdrawal†

Intravenous dosage

Adults

4 mg IV every 15 to 20 minutes for 2 doses, then 8 mg IV every 15 to 20 minutes for 2 doses, then 16 mg IV every 15 to 20 minutes for 3 doses as needed. If the patient is hyperdynamic and agitated after lorazepam 40 mg within 3 hours, consider phenobarbital or propofol.

Continuous Intravenous Infusion dosage

Adults

1 to 20 mg/hour continuous IV infusion. Titrate dose to target clinical score. Average dose: 14 mg/hour.

For chemotherapy-induced nausea/vomiting prophylaxis† as an adjunct to antiemetics

Intravenous dosage

Adults

Doses of 0.025 mg/kg IV have been reported to be effective in reducing emesis and anxiety due to chemotherapy with minimal adverse effects. Alternatively, 1.5 mg/m2 (Usual Max: 3 mg) IV can be given 45 minutes prior to initiation of chemotherapy. Dosage generally produces some amnesia of short-term memory.

Children and Adolescents

0.04 to 0.05 mg/kg IV as a single dose administered 30 minutes prior to chemotherapy. Infuse over 15 to 20 minutes. Max: 4 mg/dose. Alternatively, 0.025 to 0.05 mg/kg/dose IV every 6 hours as needed for management of anticipatory or breakthrough nausea/vomiting. Max: 4 mg/dose.

Oral dosage

Children and Adolescents

Limited data available; 0.025 to 0.05 mg/kg/dose PO every 6 hours as needed for management of anticipatory nausea/vomiting. Max: 4 mg/dose. May start 12 to 24 hours prior to chemotherapy.

For the treatment of agitation†

Oral dosage

Adults

2 mg PO every 30 to 60 minutes as needed.

Children and Adolescents

0.05 to 0.1 mg/kg/dose (Max: 2 mg/dose) PO every 30 to 60 minutes as needed.

Intramuscular dosage

Adults

2 to 4 mg IM every 30 to 60 minutes as needed.

Children and Adolescents

0.05 to 0.1 mg/kg/dose (Max: 2 mg/dose) IM every 30 to 60 minutes as needed.

Intravenous dosage

Adults

2 mg IV every 30 to 60 minutes as needed.

Children and Adolescents

0.05 to 0.1 mg/kg/dose (Max: 2 mg/dose) IV every 30 to 60 minutes as needed.

For the treatment of cocaine-associated acute coronary syndromes†, including acute myocardial infarction, NSTEMI† or unstable angina†

Intravenous dosage

Adults

1 mg IV as a single dose, initially; may repeat dose after 5 minutes if chest pain persists.

For the treatment of cyclic vomiting syndrome†

Intravenous dosage

Adults

1 to 2 mg IV as a single dose plus diphenhydramine for additional sedation. May continue lorazepam for 24 to 48 hours if initially effective and needed.

Children and Adolescents

0.05 to 0.1 mg/kg/dose (Max: 2 mg/dose) IV every 6 hours as needed.

For the treatment of vertigo† and nausea† associated with Meniere disease

Oral dosage (immediate-release)

Adults

1 to 2 mg PO every 8 hours as needed.

Intravenous or Intramuscular dosage

Adults

2 mg IV or IM every 8 hours as needed.

For the treatment of end-of-life delirium†

For the treatment of end-of-life mixed or hyperactive delirium†

Intravenous dosage

Adults

1 mg IV every 4 hours with rescue doses of 1 mg IV every 1 hour as needed, initially. Adjust dose by 1 mg/dose based on clinical response.

For the treatment of end-of-life hypoactive delirium†

Intravenous dosage

Adults

1 to 2 mg IV every 1 hour as needed.

Contraindications And Precaution

Drug Interactions

The coadministration of certain medications may lead to harm and require avoidance or therapy modification; review all drug interactions prior to concomitant use of other medications.

Hypersensitivity

This medication is contraindicated in patients with a history of hypersensitivity to it or any of its components. Lorazepam is contraindicated in people with known hypersensitivity to other benzodiazepines. Lorazepam 1 mg extended-release capsules are also contraindicated in people with tartrazine dye hypersensitivity, as they contain tartrazine (FD and C Yellow No. 5) and may cause allergic-type reactions in susceptible

patients. Lorazepam injection is contraindicated in patients who are hypersensitive to other ingredients in these products (i.e., propylene glycol or polyethylene glycol). Some formulations of lorazepam injection also contain benzyl alcohol and are contraindicated in patients with known benzyl alcohol hypersensitivity.

Treatment-Related Restrictions of Use

Oral lorazepam is not recommended in people with psychosis; however, benzodiazepines are commonly used in clinical practice for the acute management of psychosis and mania, as well as in the treatment of extrapyramidal symptoms associated with antipsychotics.
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depression, suicidal ideation

Pre-existing depression may emerge or worsen during use of benzodiazepines. Lorazepam is not recommended for use in patients with a primary depressive disorder without adequate antidepressant therapy. Appropriate precautions (e.g., limiting the total prescription size and increased monitoring for suicidal ideation) should be considered in patients with depression.

intraarterial administration

Injectable lorazepam is contraindicated for intraarterial administration due to the possibility of arteriospasm and resultant gangrene that may require amputation.

coadministration with certain medications, pulmonary disease, respiratory insufficiency, sleep apnea

Deaths have been reported in people with severe pulmonary disease after initiation of benzodiazepines; closely monitor patients with impaired respiratory function (respiratory insufficiency). Lorazepam injection is contraindicated in patients with sleep apnea syndrome or severe respiratory insufficiency who are not receiving mechanical ventilation. If signs and symptoms of respiratory depression, hypoventilation, or apnea occur, discontinue lorazepam. Lorazepam coadministration with certain medications, such as opioids, may result in serious adverse effects, including profound sedation, CNS depression, respiratory depression, coma, and death. The concomitant use of these medications has demonstrated an increased risk of drug-related mortality compared to the use of opioids alone. The combined use of opioids and benzodiazepines should be reserved for patients in whom alternative treatment options are inadequate. If the decision is made to prescribe lorazepam concomitantly with opioids, the lowest effective dose should be prescribed for the shortest duration of use, and patients should be followed closely for signs and symptoms of respiratory depression and sedation. Advise

patients and caregivers about the risks of respiratory depression and sedation when lorazepam is used with opioids. Due to the risks associated with concomitant use, the ASAM clinical practice guidelines suggest re-evaluation of the risks and benefits of continued benzodiazepine prescribing in patients taking opioids at least every 3 months or with every related clinical encounter or prescription renewal, whichever is more frequent.

abrupt discontinuation, increased risk for benzodiazepine withdrawal, substance abuse disorder

The use of benzodiazepines, including lorazepam, exposes users to serious adverse events, including risks of abuse, misuse, and addiction, which can lead to overdose or death. Abuse and misuse of benzodiazepines commonly involve concomitant use of other medications, alcohol, and/or illicit substances, which can increase the frequency of serious adverse outcomes. Before prescribing and throughout treatment, assess each patient's risk for abuse, misuse, and addiction using a standardized screening tool. Counsel patients about the risks and proper use of lorazepam and prescribe the lowest effective dose to reduce excess exposure and advise patients on proper disposal of unused medication. Minimize the concomitant use of CNS depressants or other substances associated with abuse, misuse, and addiction (e.g., opioid analgesics, stimulants). If a substance abuse disorder is suspected, evaluate the patient and institute or refer them for early treatment. Chronic, continued use of benzodiazepines may also lead to physical benzodiazepine dependence. The risk of dependence increases with longer treatment duration and higher daily doses and may lead to an increased risk for benzodiazepine withdrawal upon discontinuation. Abrupt discontinuation or rapid dose reduction can precipitate acute withdrawal reactions, which can be life-threatening. To reduce the risk of withdrawal reactions, use a gradual, patient-specific taper to reduce the dose or discontinue lorazepam. In some cases, patients using benzodiazepines have developed a protracted withdrawal syndrome with symptoms lasting weeks to more than 12 months. Individuals with a history of a seizure disorder should not be withdrawn abruptly from benzodiazepines due to the risk of precipitating seizures; status epilepticus has also been reported. Clinicians should be aware that the use of flumazenil may increase the risk of seizures, particularly in long-term users of benzodiazepines.

activities requiring coordination and concentration, driving or operating machinery

Particular caution is required in determining the amount of time needed after outpatient procedures or surgery before it is safe for any patient to ambulate. No individual should get out of bed unassisted within 8 hours of lorazepam injection. In

addition, patients should not attempt driving or operating machinery until 24 to 48 hours after surgery or until the CNS depressant effects have subsided, whichever is longer. Due to CNS depressive effects, individuals should be cautioned against driving or operating machinery or participating in other activities requiring coordination and concentration until they know how lorazepam may affect them. Some patients may experience excessive sedation and impaired ability to perform tasks. Given the primary CNS effects of lorazepam, use caution during coadministration with other CNS depressants. Alcohol may further impair cognitive and motor skills and can increase the risk of serious CNS or respiratory depression; alcohol use should be avoided during lorazepam treatment.

renal failure, renal impairment

Lorazepam is excreted renally as an inactive metabolite; less than 1% is excreted unchanged. No specific dosage adjustments are recommended for renal impairment or renal failure. However, due to lack of data especially in patients with kidney failure, it is advisable to start with the lowest dosage and titrate to effectiveness and tolerance and monitor closely for excessive sedation or other adverse effects.

acute angle-closure glaucoma

Injectable and oral lorazepam formulations are contraindicated in individuals with acute angle-closure glaucoma as they may exacerbate narrow-angle closure.

myasthenia gravis

Lorazepam should be used with caution in patients with myasthenia gravis, particularly at higher doses. These patients may be more sensitive to the CNS and respiratory effects of the benzodiazepines.

hepatic encephalopathy, hepatic failure

Lorazepam can be considered when a benzodiazepine is required in patients with hepatic failure due to the low hepatic extraction, glucuronidation as the primary metabolic pathway, and lack of active metabolites. Initiate with lower dosages and carefully monitor for sedation and other adverse effects. Dosage for patients with severe hepatic failure should be adjusted carefully according to patient response; lower doses may be sufficient in such patients. As with all benzodiazepines, the use of lorazepam may worsen hepatic encephalopathy; therefore, lorazepam should be used with caution in patients with severe hepatic insufficiency and/or hepatic encephalopathy.

geriatric

Lorazepam should be used cautiously in geriatric adults since they are more sensitive to the effects of benzodiazepines; lower initial doses and slower titration are recommended. Because lorazepam can cause drowsiness and a decreased level of consciousness, there is a higher risk of falls, particularly in the elderly, with the potential for subsequent severe injuries. The Beers Criteria state that benzodiazepines are potentially inappropriate medications in geriatric adults with a history of delirium, dementia, or a history of falls or fractures; benzodiazepines increase the risk of cognitive impairment, delirium, falls, fractures, and motor vehicle accidents in older adults and avoidance is generally recommended, with some exceptions for severe generalized anxiety disorder, seizures, and other approved indications. If a benzodiazepine must be used, consider reducing the use of other CNS-active medications and implement other strategies to reduce fall risk. The Beers Criteria are not meant to apply to patients at the end of life or receiving palliative care, when risk-benefit considerations of drug therapy can be different. The U.S. Omnibus Budget Reconciliation Act (OBRA) regulates anxiolytic and sedative/hypnotic use in long-term care facilities. When a drug is used to induce sleep, treat a sleep disorder, manage behavior, stabilize mood, or treat a psychiatric disorder, tapering as outlined in the OBRA guidelines should be attempted unless clinically contraindicated. Dosages and durations of treatment used should align with prescribing labels, published literature recommendations, and expert guidelines.

children, infants, neonates, premature neonates

Lorazepam injection is contraindicated in premature neonates. Both lorazepam oral solution concentrated and injectable lorazepam solutions contain propylene glycol and polyethylene glycol. Lorazepam injection also contains benzyl alcohol as a preservative. Pediatric patients, in particular neonates, may be more sensitive to these compounds. Although normal therapeutic doses of lorazepam contain very small amounts of propylene glycol, polyethylene glycol, and benzyl alcohol, the clinician should be aware of the toxic potential, especially if other drugs containing the compounds are administered. Excessive propylene glycol can cause lactic acidosis, hyperosmolality, tachypnea, tachycardia, diaphoresis, and central nervous system toxicity (e.g., seizures, intraventricular hemorrhage). Excessive amounts of benzyl alcohol in neonates have been associated with hypotension, metabolic acidosis, and kernicterus. A "gasping syndrome" characterized by CNS depression, metabolic acidosis, and gasping respirations has been associated with benzyl alcohol dosages more than 99 mg/kg/day in neonates. However, the minimum amount of benzyl alcohol at which toxicity may occur is unknown, and premature and low-birth-weight neonates may be more likely to develop toxicity. Repeated or lengthy use of general anesthetic and sedation drugs

during surgeries or procedures in neonates, infants, and children younger than 3 years of age, including in utero exposure, may have negative effects on brain development. Consider the benefits of appropriate anesthesia in young children against the potential risks, especially for procedures that may last more than 3 hours or if multiple procedures are required during the first 3 years of life. It may be appropriate to delay certain procedures if doing so will not jeopardize the health of the child. No specific anesthetic or sedation drug has been shown to be safer than another. Human studies suggest that a single short exposure to a general anesthetic in young pediatric patients is unlikely to have negative effects on behavior and learning; however, further research is needed to fully characterize how anesthetic exposure affects brain development. Although commonly used off-label in the pediatric population, safe and effective use of immediate-release oral and parenteral lorazepam has not been established in pediatric patients younger than 12 years and 18 years, respectively. The safety and efficacy of lorazepam extended-release capsules have not been established in pediatric patients.

labor, obstetric delivery, pregnancy

Published data from observational studies of benzodiazepine use during pregnancy have not established a drug-associated risk of major birth defects. Early studies suggested an increased risk of congenital malformations with diazepam and chlordiazepoxide use. However, more recent case-control and cohort studies of benzodiazepine use during pregnancy, adjusted for confounding exposures to other substances, have not confirmed these findings. Benzodiazepines cross the placenta. Neonates exposed to benzodiazepines during late pregnancy, including during labor and delivery, may experience symptoms of sedation. Those with prolonged exposure during late pregnancy may experience withdrawal after delivery. The incidence, time to onset, and duration of sedation and withdrawal are multifactorial (e.g., duration of use, drug lipophilicity, placental disposition, degree of accumulation in neonatal tissues). Individuals who become pregnant or intend to become pregnant while taking lorazepam should be advised to discuss the possibility of discontinuing the drug with their physician. If a benzodiazepine is required during pregnancy, avoid first trimester administration if possible, consider short-acting agents, limit treatment to the shortest possible duration and lowest effective dose, and discontinue the drug well before delivery. There are insufficient data about the obstetrical safety of parenteral lorazepam. Use during labor and obstetric delivery is not recommended. Repeated or lengthy use of general anesthetic and sedation drugs during surgeries or procedures during the third trimester of pregnancy may have negative effects on fetal brain development. Consider the benefits of appropriate anesthesia in pregnant people against the potential risks, especially for procedures that may last more than 3 hours or if multiple procedures are required prior to delivery. It may be appropriate to delay certain procedures if doing so will not jeopardize the health of the child and/or pregnant person. No specific anesthetic

or sedation drug has been shown to be safer than another. Human studies suggest that a single short exposure to a general anesthetic in young pediatric patients is unlikely to have negative effects on behavior and learning; however, further research is needed to fully characterize how anesthetic exposure affects brain development. There is a pregnancy registry that monitors pregnancy outcomes in pregnant people exposed to lorazepam or other psychiatric medications during pregnancy. Healthcare providers are encouraged to register patients by calling the National Pregnancy Registry for Psychiatric Medications at 1-866-961-2388 or visiting online at https://womensmentalhealth.org/research/pregnancyregistry/.

neonates and infants exposed to this medication in utero

Monitor neonates and infants exposed to this medication in utero for symptoms of withdrawal, such as hyperreflexia, irritability, restlessness, tremors, inconsolable crying, and feeding difficulties. In neonates exposed during late pregnancy, including during labor and delivery, monitor for sedation, including respiratory depression, lethargy, and hypotonia. If symptoms are present, manage accordingly.

breast-feeding

Use lorazepam with caution during breast-feeding. Lorazepam is excreted into human milk in low concentrations. Although lorazepam is often considered a preferred benzodiazepine during lactation due to its shorter half-life and lack of active metabolite, there is potential for serious adverse reactions in the breastfed child. Neonatal metabolism of lorazepam is significantly slower compared to adults. If used during lactation, prescribe lorazepam at the lowest effective dosage and duration. Monitor exposed children for excessive sedation, slowed breathing rate, poor feeding, and inadequate weight gain. Withdrawal may occur in those with long-term exposure to the drug. This may occur with cessation of breast-feeding or cessation of the medication in the breast-feeding person. In a study of 4 lactating people, concentrations of free lorazepam in breast milk 4 hours after a single 3.5 mg oral dose were found to be 8 to 9 nanograms/mL, which accounted for 14.8% to 25.7% of the lactating person's plasma concentration. In a separate report, a person taking lorazepam 2.5 mg by mouth twice daily for the first 5 days postpartum had milk concentrations of free and conjugated lorazepam of 12 and 35 mcg/L, respectively, at an unspecified time on day 5, and their infant showed no signs of sedation. In a retrospective cohort study of breast-feeding people using a benzodiazepine (n = 124), sedation was not reported in any infant exposed to lorazepam through breast milk (52% of participants).

Pregnancy And Lactation

Use lorazepam with caution during breast-feeding. Lorazepam is excreted into human milk in low concentrations. Although lorazepam is often considered a preferred benzodiazepine during lactation due to its shorter half-life and lack of active metabolite, there is potential for serious adverse reactions in the breastfed child. Neonatal metabolism of lorazepam is significantly slower compared to adults. If used during lactation, prescribe lorazepam at the lowest effective dosage and duration. Monitor exposed children for excessive sedation, slowed breathing rate, poor feeding, and inadequate weight gain. Withdrawal may occur in those with long-term exposure to the drug. This may occur with cessation of breast-feeding or cessation of the medication in the breast-feeding person. In a study of 4 lactating people, concentrations of free lorazepam in breast milk 4 hours after a single 3.5 mg oral dose were found to be 8 to 9 nanograms/mL, which accounted for 14.8% to 25.7% of the lactating person's plasma concentration. In a separate report, a person taking lorazepam 2.5 mg by mouth twice daily for the first 5 days postpartum had milk concentrations of free and conjugated lorazepam of 12 and 35 mcg/L, respectively, at an unspecified time on day 5, and their infant showed no signs of sedation. In a retrospective cohort study of breast-feeding people using a benzodiazepine (n = 124), sedation was not reported in any infant exposed to lorazepam through breast milk (52% of participants).

Interactions

Acetaminophen; Aspirin, ASA; Caffeine: (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Acetaminophen; Aspirin; diphenhydrAMINE: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Acetaminophen; Caffeine: (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Acetaminophen; Caffeine; Dihydrocodeine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and

sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines. (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Acetaminophen; Caffeine; Pyrilamine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Acetaminophen; Chlorpheniramine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Acetaminophen; Chlorpheniramine; Dextromethorphan: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Acetaminophen; Chlorpheniramine; Dextromethorphan; Phenylephrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Acetaminophen; Chlorpheniramine; Dextromethorphan; Pseudoephedrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Acetaminophen; Chlorpheniramine; Phenylephrine : (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Acetaminophen; Codeine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking

benzodiazepines.
Acetaminophen; Dextromethorphan; Doxylamine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Acetaminophen; Dextromethorphan; guaiFENesin; Phenylephrine: (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Acetaminophen; Dextromethorphan; Phenylephrine: (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Acetaminophen; diphenhydrAMINE: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Acetaminophen; guaiFENesin; Phenylephrine: (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Acetaminophen; HYDROcodone: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If hydrocodone is initiated in a patient taking a benzodiazepine, reduce initial dosage and titrate to clinical response; for hydrocodone extended-release products, initiate hydrocodone at 20% to 30% of the usual dosage. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid opiate cough medications in patients taking benzodiazepines.
Acetaminophen; oxyCODONE: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If oxycodone is initiated in a patient taking a benzodiazepine, reduce dosages and titrate to clinical response. For acetaminophen; oxycodone extended-release tablets, start with 1 tablet PO every 12 hours, and for other oxycodone products, use an initial dose of oxycodone at 1/3 to 1/2 the usual dosage. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical

response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Acetaminophen; Pamabrom; Pyrilamine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Acetaminophen; Phenylephrine: (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
ALFentanil: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
ALPRAZolam: (Moderate) Concomitant administration of alprazolam with CNS-depressant drugs, such as lorazepam, can potentiate the CNS effects of either agent.
Amitriptyline: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
Amobarbital: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use.
Amoxapine: (Moderate) Amoxapine may enhance the response to the effects of benzodiazepines and other CNS depressants. Patients should be warned of the possibility of drowsiness that may impair performance of potentially hazardous tasks such as driving an automobile or operating machinery.
Apomorphine: (Moderate) Apomorphine causes significant somnolence. Concomitant administration of apomorphine and benzodiazepines could result in additive depressant effects.
Apraclonidine: (Minor) No specific drug interactions were identified with systemic agents and apraclonidine during clinical trials. Theoretically, apraclonidine might potentiate the effects of CNS depressant drugs such as the anxiolytics, sedatives, and hypnotics, including barbiturates or benzodiazepines.
ARIPiprazole: (Moderate) Monitor blood pressure and for unusual drowsiness and sedation during coadministration of aripiprazole and benzodiazepines. Intensity of sedation and orthostatic hypotension were greater with the combination of oral aripiprazole and lorazepam compared to aripiprazole alone.

Asenapine: (Moderate) Drugs that can cause CNS depression, if used concomitantly with asenapine, may increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, and dizziness. Caution should be used when asenapine is given in combination with other centrally-acting medications including anxiolytics, sedatives, and hypnotics (including barbiturates), buprenorphine, buprenorphine; naloxone, butorphanol, dronabinol, THC, nabilone, nalbuphine, opiate agonists, pentazocine, acetaminophen; pentazocine, aspirin, ASA; pentazocine, and pentazocine; naloxone.
Aspirin, ASA; Butalbital; Caffeine: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use. (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Aspirin, ASA; Caffeine: (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Aspirin, ASA; Caffeine; Orphenadrine: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation. (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Aspirin, ASA; Carisoprodol; Codeine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines. (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.
Aspirin, ASA; oxyCODONE: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is

necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If oxycodone is initiated in a patient taking a benzodiazepine, reduce dosages and titrate to clinical response. For acetaminophen; oxycodone extended-release tablets, start with 1 tablet PO every 12 hours, and for other oxycodone products, use an initial dose of oxycodone at 1/3 to 1/2 the usual dosage. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Atazanavir: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and atazanavir is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and atazanavir is an UGT inhibitor.
Atazanavir; Cobicistat: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and atazanavir is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and atazanavir is an UGT inhibitor.
Atropine; Difenoxin: (Moderate) Concomitant administration of benzodiazepines with CNS-depressant drugs, such as diphenoxylate/difenoxin, can potentiate the CNS effects of either agent.
Azelastine: (Moderate) Monitor for excessive sedation and somnolence during coadministration of azelastine and benzodiazepines. Concurrent use may result in additive CNS depression.
Azelastine; Fluticasone: (Moderate) Monitor for excessive sedation and somnolence during coadministration of azelastine and benzodiazepines. Concurrent use may result in additive CNS depression.
Baclofen: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.
Barbiturates: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use.
Belladonna; Opium: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a

lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Belumosudil: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and belumosudil is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and belumosudil is an UGT inhibitor.
Benzhydrocodone; Acetaminophen: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If benzhydrocodone is initiated in a patient taking a benzodiazepine, reduce initial dosage and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Benztropine: (Moderate) CNS depressants, such as anxiolytics, sedatives, and hypnotics, can increase the sedative effects of benztropine.
Brexanolone: (Moderate) Concomitant use of brexanolone with CNS depressants like the benzodiazepines may increase the likelihood or severity of adverse reactions related to sedation and additive CNS depression. Monitor for excessive sedation, dizziness, and a potential for loss of consciousness during brexanolone use.
Brimonidine: (Moderate) Based on the sedative effects of brimonidine in individual patients, brimonidine administration has potential to enhance the CNS depressants effects of the anxiolytics, sedatives, and hypnotics including benzodiazepines.
Brimonidine; Brinzolamide: (Moderate) Based on the sedative effects of brimonidine in individual patients, brimonidine administration has potential to enhance the CNS depressants effects of the anxiolytics, sedatives, and hypnotics including benzodiazepines.
Brimonidine; Timolol: (Moderate) Based on the sedative effects of brimonidine in individual patients, brimonidine administration has potential to enhance the CNS depressants effects of the anxiolytics, sedatives, and hypnotics including benzodiazepines.
Brompheniramine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.

Brompheniramine; Dextromethorphan; Phenylephrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Brompheniramine; Phenylephrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Brompheniramine; Pseudoephedrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Brompheniramine; Pseudoephedrine; Dextromethorphan: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Buprenorphine: (Major) Concomitant use of mixed opiate agonists/antagonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of mixed opiate agonists/antagonists with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If a mixed opiate agonist/antagonist is initiated for pain in a patient taking a benzodiazepine, use a lower initial dose of the mixed opiate agonist/antagonist and titrate to clinical response. Reduce injectable buprenorphine dose by 1/2, and for the buprenorphine transdermal patch, start therapy with the 5 mcg/hour patch. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking a mixed opiate agonist/antagonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. In patients treated with buprenorphine for opioid use disorder, cessation of benzodiazepines or other CNS depressants is preferred in most cases. Consider alternatives to benzodiazepines for conditions such as anxiety or insomnia in patients receiving buprenorphine maintenance treatment. Educate patients about the risks and symptoms of respiratory depression and sedation.
Buprenorphine; Naloxone: (Major) Concomitant use of mixed opiate agonists/antagonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of mixed opiate agonists/antagonists with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If a mixed opiate agonist/antagonist is initiated for pain in a patient taking a benzodiazepine, use a lower initial dose of the mixed opiate agonist/antagonist and

titrate to clinical response. Reduce injectable buprenorphine dose by 1/2, and for the buprenorphine transdermal patch, start therapy with the 5 mcg/hour patch. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking a mixed opiate agonist/antagonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. In patients treated with buprenorphine for opioid use disorder, cessation of benzodiazepines or other CNS depressants is preferred in most cases. Consider alternatives to benzodiazepines for conditions such as anxiety or insomnia in patients receiving buprenorphine maintenance treatment. Educate patients about the risks and symptoms of respiratory depression and sedation.
Butalbital; Acetaminophen: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use.
Butalbital; Acetaminophen; Caffeine: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use. (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Butalbital; Acetaminophen; Caffeine; Codeine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines. (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use. (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Butalbital; Aspirin; Caffeine; Codeine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical

response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines. (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use. (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Butorphanol: (Major) Concomitant use of mixed opiate agonists/antagonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of mixed opiate agonists/antagonists with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If a mixed opiate agonist/antagonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the mixed opiate agonist/antagonist and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking a mixed opiate agonist/antagonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Caffeine: (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Caffeine; Sodium Benzoate: (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Calcium, Magnesium, Potassium, Sodium Oxybates: (Contraindicated) Sodium oxybate should not be used in combination with CNS depressant anxiolytics, sedatives, and hypnotics or other sedative CNS depressant drugs. Specifically, sodium oxybate use is contraindicated in patients being treated with sedative hypnotic drugs. Sodium oxybate (GHB) has the potential to impair cognitive and motor skills. For example, the concomitant use of barbiturates and benzodiazepines increases sleep duration and may contribute to rapid onset, pronounced CNS depression, respiratory depression, or coma when combined with sodium oxybate.
Cannabidiol: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and cannabidiol is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is a UGT2B7 substrate. In vitro data predicts inhibition of UGT2B7 by cannabidiol, potentially resulting in clinically significant interactions.
Capivasertib: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and capivasertib is necessary. Avoid lorazepam extended-release capsules and utilize

lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and capivasertib is an UGT inhibitor.
Capsaicin; Metaxalone: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.
Carbachol; Brimonidine: (Moderate) Based on the sedative effects of brimonidine in individual patients, brimonidine administration has potential to enhance the CNS depressants effects of the anxiolytics, sedatives, and hypnotics including benzodiazepines.
Carbidopa; Levodopa: (Moderate) Advise patients of possible additive risk for somnolence during concomitant use of levodopa with sedating agents, such as benzodiazepines. Concomitant use may also increase the risk for sudden sleep onset during activities of daily living, such as driving or eating; this adverse effect may occur without any warning signs. Patients should consider avoiding activities such as driving or operating machinery until the effects of this combination are known.
Carbidopa; Levodopa; Entacapone: (Major) Concomitant administration of benzodiazepines with other drugs have CNS depressant properties, including COMT inhibitors, can potentiate the CNS effects of either agent. COMT inhibitors have also been associated with sudden sleep onset during activities of daily living such as driving, which has resulted in accidents in some cases. Prescribers should re-assess patients for drowsiness or sleepiness regularly throughout treatment, especially since events may occur well after the start of treatment. Patients should be advised to avoid driving or other tasks requiring mental alertness until they know how the combination affects them. (Moderate) Advise patients of possible additive risk for somnolence during concomitant use of levodopa with sedating agents, such as benzodiazepines. Concomitant use may also increase the risk for sudden sleep onset during activities of daily living, such as driving or eating; this adverse effect may occur without any warning signs. Patients should consider avoiding activities such as driving or operating machinery until the effects of this combination are known.
Carbinoxamine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Cariprazine: (Moderate) Due to the CNS effects of cariprazine, caution should be used when cariprazine is given in combination with other centrally-acting medications including benzodiazepines and other anxiolytics, sedatives, and hypnotics.
Carisoprodol: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.

Celecoxib; Tramadol: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Cenobamate: (Moderate) Monitor for excessive sedation and somnolence during coadministration of cenobamate and benzodiazepines. Concurrent use may result in additive CNS depression.
Cetirizine: (Moderate) Concurrent use of cetirizine/levocetirizine with benzodiazepines should generally be avoided. Coadministration may increase the risk of CNS depressant-related side effects. If concurrent use is necessary, monitor for excessive sedation and somnolence.
Cetirizine; Pseudoephedrine: (Moderate) Concurrent use of cetirizine/levocetirizine with benzodiazepines should generally be avoided. Coadministration may increase the risk of CNS depressant-related side effects. If concurrent use is necessary, monitor for excessive sedation and somnolence.
Chlophedianol; Dexbrompheniramine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Chlophedianol; Dexchlorpheniramine; Pseudoephedrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Chlorcyclizine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
chlordiazePOXIDE; Amitriptyline: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
Chlorpheniramine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Chlorpheniramine; Codeine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is

necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines. (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Chlorpheniramine; Dextromethorphan: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Chlorpheniramine; Dextromethorphan; Phenylephrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Chlorpheniramine; Dextromethorphan; Pseudoephedrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Chlorpheniramine; HYDROcodone: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If hydrocodone is initiated in a patient taking a benzodiazepine, reduce initial dosage and titrate to clinical response; for hydrocodone extended-release products, initiate hydrocodone at 20% to 30% of the usual dosage. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid opiate cough medications in patients taking benzodiazepines. (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Chlorpheniramine; Ibuprofen; Pseudoephedrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Chlorpheniramine; Phenylephrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution

with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Chlorpheniramine; Pseudoephedrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
chlorproMAZINE: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Chlorzoxazone: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.
Cisapride: (Moderate) Cisapride may enhance the sedative effects of benzodiazepines. Patients should not drive or operate heavy machinery until they know how the combination affects them. Patient counseling is important, as cisapride alone does not cause drowsiness or affect psychomotor function.
Clemastine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
cloBAZam: (Major) Use clobazam with other benzodiazepines with caution due to the risk for additive CNS depression.
clomiPRAMINE: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
cloNIDine: (Moderate) Clonidine has CNS depressive effects and can potentiate the actions of other CNS depressants including benzodiazepines.
cloZAPine: (Moderate) If concurrent therapy with clozapine and a benzodiazepine is necessary, it is advisable to begin with the lowest possible benzodiazepine dose and closely monitor the patient, particularly at initiation of treatment and following dose increases. Although the combination has been used safely, adverse reactions such as confusion, ataxia, somnolence, delirium, collapse, cardiac arrest, respiratory arrest, and death have occurred rarely in patients receiving clozapine concurrently or following benzodiazepine therapy. Several benzodiazepines, including clonazepam, oxazepam, flurazepam, diazepam, clobazam, flunitrazepam, and lorazepam have been implicated in these reactions. At least one case of sudden death was reported following intravenous administration of lorazepam to a patient receiving clozapine.
Codeine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest

effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines.
Codeine; Dexbrompheniramine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines. (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Codeine; guaiFENesin: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines.
Codeine; guaiFENesin; Pseudoephedrine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an

opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines.
Codeine; Phenylephrine; Promethazine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines. (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Codeine; Promethazine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid prescribing opiate cough medications in patients taking benzodiazepines. (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Colesevelam: (Moderate) Colesevelam may decrease the absorption of anticonvulsants. To minimize potential for interactions, consider administering oral anticonvulsants at least 1 hour before or at least 4 hours after colesevelam.
COMT inhibitors: (Major) Concomitant administration of benzodiazepines with other drugs have CNS depressant properties, including COMT inhibitors, can potentiate the CNS effects of either agent. COMT inhibitors have also been associated with sudden

sleep onset during activities of daily living such as driving, which has resulted in accidents in some cases. Prescribers should re-assess patients for drowsiness or sleepiness regularly throughout treatment, especially since events may occur well after the start of treatment. Patients should be advised to avoid driving or other tasks requiring mental alertness until they know how the combination affects them.
Cyclobenzaprine: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.
Cyproheptadine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Dantrolene: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.
Daridorexant: (Major) Monitor for excessive sedation and somnolence during use of daridorexant with lorazepam. Dosage adjustments may be necessary when administered together because of potentially additive CNS effects. Use of more than 2 hypnotics should be avoided due to the additive CNS depressant and complex sleep-related behaviors that may occur. While anxiolytic medications may be used concurrently with daridorexant, a reduction in dose of one or both agents may be needed. The risk of next-day impairment, including impaired driving, is increased if daridorexant is taken with other CNS depressants.
Desflurane: (Moderate) Concurrent use with benzodiazepines can decrease the minimum alveolar concentration (MAC) of desflurane needed to produce anesthesia.
Desipramine: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
Desogestrel: Ethinyl Estradiol; Ferrous Bisglycinate: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Desogestrel; Ethinyl Estradiol: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Deutetrabenazine: (Moderate) Advise patients that concurrent use of deutetrabenazine and drugs that can cause CNS depression, such as lorazepam, may have additive effects and worsen drowsiness or sedation.

Dexbrompheniramine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Dexbrompheniramine; Dextromethorphan; Phenylephrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Dexbrompheniramine; Pseudoephedrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Dexchlorpheniramine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Dexchlorpheniramine; Dextromethorphan; Pseudoephedrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
dexmedeTOMIDine: (Moderate) Concurrent use of dexmedetomidine and benzodiazepines may result in additive CNS depression. A reduction in dosage of dexmedetomidine or the benzodiazepine may be required.
Dextromethorphan; diphenhydrAMINE; Phenylephrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Dextromethorphan; guaiFENesin; Phenylephrine: (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Dichlorphenamide: (Moderate) Use dichlorphenamide and lorazepam together with caution. Metabolic acidosis is associated with the use of dichlorphenamide and has been reported rarely with the use of lorazepam injection for the treatment of status epilepticus. Concurrent use may increase the severity of metabolic acidosis. Measure sodium bicarbonate concentrations at baseline and periodically during dichlorphenamide treatment. If metabolic acidosis occurs or persists, consider reducing the dose or discontinuing dichlorphenamide therapy.
Dicyclomine: (Moderate) Dicyclomine can cause drowsiness, so it should be used cautiously in patients receiving CNS depressants like benzodiazepines.
Difelikefalin: (Moderate) Monitor for dizziness, somnolence, mental status changes, and gait disturbances if concomitant use of difelikefalin with CNS depressants is necessary. Concomitant use may increase the risk for these adverse reactions.

dimenhyDRINATE: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
diphenhydrAMINE: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
diphenhydrAMINE; Ibuprofen: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
diphenhydrAMINE; Naproxen: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
diphenhydrAMINE; Phenylephrine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Diphenoxylate; Atropine: (Moderate) Concomitant administration of benzodiazepines with CNS-depressant drugs, such as diphenoxylate/difenoxin, can potentiate the CNS effects of either agent.
Doxepin: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
Doxylamine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Doxylamine; Pyridoxine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
droNABinol: (Moderate) Use caution if the use of benzodiazepines are necessary with dronabinol, and monitor for additive dizziness, confusion, somnolence, and other CNS effects.
droPERidol: (Major) Droperidol administration is associated with an established risk for QT prolongation and torsades de pointes. In December 2001, the FDA issued a black box warning regarding the use of droperidol and its association with QT prolongation and potential for cardiac arrhythmias based on post-marketing surveillance data. Risk factors for the development of prolonged QT syndrome may include the use of benzodiazepines. Also, droperidol and benzodiazepines can both cause CNS depression. If used with a benzodiazepine, droperidol should be initiated at a low dose and adjusted upward, with caution, as needed to achieve the desired effect.
Drospirenone; Ethinyl Estradiol: (Minor) Ethinyl estradiol may enhance the metabolism

of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Drospirenone; Ethinyl Estradiol; Levomefolate: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Entacapone: (Major) Concomitant administration of benzodiazepines with other drugs have CNS depressant properties, including COMT inhibitors, can potentiate the CNS effects of either agent. COMT inhibitors have also been associated with sudden sleep onset during activities of daily living such as driving, which has resulted in accidents in some cases. Prescribers should re-assess patients for drowsiness or sleepiness regularly throughout treatment, especially since events may occur well after the start of treatment. Patients should be advised to avoid driving or other tasks requiring mental alertness until they know how the combination affects them.
Ergotamine; Caffeine: (Minor) Patients taking benzodiazepines for insomnia should not use caffeine-containing products prior to going to bed as these products may antagonize the sedative effects of the benzodiazepine.
Erlotinib: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and erlotinib is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and erlotinib is an UGT inhibitor.
Esketamine: (Major) Closely monitor patients receiving esketamine and benzodiazepines for sedation and other CNS depressant effects. Instruct patients who receive a dose of esketamine not to drive or engage in other activities requiring alertness until the next day after a restful sleep.
Eszopiclone: (Moderate) Concomitant administration of benzodiazepines with eszopiclone can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. The concurrent use of eszopiclone with other anxiolytics, sedatives, and hypnotics at bedtime or in the middle of the night is not recommended. In addition, the risk of next-day psychomotor impairment is increased during co-administration of eszopiclone and other CNS depressants, which may decrease the ability to perform tasks requiring full mental alertness such as driving. If used together, a reduction in the dose of one or both drugs may be needed.
Ethanol: (Major) Advise patients to avoid alcohol consumption while taking CNS depressants. Alcohol consumption may result in additive CNS depression. Alcohol may also increase drug exposure and the risk for overdose by disrupting extended-release lorazepam capsules.
Ethinyl Estradiol; Norelgestromin: (Minor) Ethinyl estradiol may enhance the metabolism

of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Ethinyl Estradiol; Norethindrone Acetate: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Ethinyl Estradiol; Norgestrel: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Ethynodiol Diacetate; Ethinyl Estradiol: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Etomidate: (Moderate) Concomitant administration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent.
Etonogestrel; Ethinyl Estradiol: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Fenfluramine: (Moderate) Monitor for excessive sedation and somnolence during coadministration of fenfluramine and benzodiazepines. Concurrent use may result in additive CNS depression.
fentaNYL: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Flumazenil: (Major) Flumazenil competes with benzodiazepines for binding at the GABA/benzodiazepine-receptor complex, the specific binding site of benzodiazepines. Because binding at the receptor is competitive and flumazenil has a much shorter duration of action than do most benzodiazepines, it is possible for the effects of flumazenil to dissipate sooner than the effects of the benzodiazepine. Flumazenil does not affect the pharmacokinetics of the benzodiazepines. Abrupt awakening can cause

dysphoria, agitation, and possibly increased adverse effects. If administered to patients who have received a benzodiazepine chronically, abrupt interruption of benzodiazepine agonism by flumazenil can induce benzodiazepine withdrawal including seizures. Flumazenil has minimal effects on benzodiazepine-induced respiratory depression; suitable ventilatory support should be available, especially in treating acute benzodiazepine overdose. Flumazenil does not reverse the actions of barbiturates, opiate agonists, or tricyclic antidepressants.
fluPHENAZine: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Food: (Major) Advise patients to avoid cannabis use while taking CNS depressants due to the risk for additive CNS depression and potential for other cognitive adverse reactions.
Foscarbidopa; Foslevodopa: (Moderate) Advise patients of possible additive risk for somnolence during concomitant use of levodopa with sedating agents, such as benzodiazepines. Concomitant use may also increase the risk for sudden sleep onset during activities of daily living, such as driving or eating; this adverse effect may occur without any warning signs. Patients should consider avoiding activities such as driving or operating machinery until the effects of this combination are known.
Gabapentin: (Major) Concomitant use of benzodiazepines with gabapentin may cause excessive sedation, somnolence, and respiratory depression. If concurrent use is necessary, initiate gabapentin at the lowest recommended dose and monitor patients for symptoms of respiratory depression and sedation. Educate patients about the risks and symptoms of excessive CNS depression and respiratory depression.
Gemfibrozil: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and gemfibrozil is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and gemfibrozil is an UGT inhibitor.
General anesthetics: (Moderate) Concomitant administration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent.
Glecaprevir; Pibrentasvir: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and glecaprevir is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and glecaprevir is an UGT inhibitor. (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and pibrentasvir is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and pibrentasvir is an UGT inhibitor.

guaiFENesin; Phenylephrine: (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
guanFACINE: (Moderate) Guanfacine has been associated with sedative effects and can potentiate the actions of other CNS depressants including benzodiazepines.
Haloperidol: (Moderate) Haloperidol can potentiate the actions of other CNS depressants, such as benzodiazepines, Caution should be exercised with simultaneous use of these agents due to potential excessive CNS effects.
Homatropine; HYDROcodone: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If hydrocodone is initiated in a patient taking a benzodiazepine, reduce initial dosage and titrate to clinical response; for hydrocodone extended-release products, initiate hydrocodone at 20% to 30% of the usual dosage. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid opiate cough medications in patients taking benzodiazepines.
HYDROcodone: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If hydrocodone is initiated in a patient taking a benzodiazepine, reduce initial dosage and titrate to clinical response; for hydrocodone extended-release products, initiate hydrocodone at 20% to 30% of the usual dosage. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid opiate cough medications in patients taking benzodiazepines.
HYDROcodone; Ibuprofen: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If hydrocodone is initiated in a patient taking a benzodiazepine, reduce initial dosage and titrate to clinical response; for hydrocodone extended-release products, initiate hydrocodone at 20% to 30% of the usual dosage. If a

benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation. Avoid opiate cough medications in patients taking benzodiazepines.
HYDROmorphone: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If hydromorphone is initiated in a patient taking a benzodiazepine, reduce the initial dosage of hydromorphone and titrate to clinical response; for hydromorphone extended-release tablets, use 1/3 to 1/2 of the estimated hydromorphone starting dose. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Hydroxychloroquine: (Moderate) Monitor persons with epilepsy for seizure activity during concomitant lorazepam and hydroxychloroquine use. Hydroxychloroquine can lower the seizure threshold; therefore, the activity of antiepileptic drugs may be impaired with concomitant use.
hydrOXYzine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Iloperidone: (Moderate) Drugs that can cause CNS depression, if used concomitantly with iloperidone, may increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, and dizziness. Caution should be used when iloperidone is given in combination with other centrally-acting medications including anxiolytics, sedatives, and hypnotics.
Imipramine: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
Iopamidol: (Moderate) The use of intrathecal radiopaque contrast agents is associated with a risk of seizures. Patients should be instructed to continue using benzodiazepines during procedures or exams that require the use of intrathecal radiopaque contrast agents as abrupt discontinuation of benzodiazepines may also increase seizure risk.
Isocarboxazid: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of benzodiazepines and monoamine oxidase inhibitors (MAOIs) due to the risk for additive CNS depression.
Isoflurane: (Moderate) Concomitant administration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent.

Ketamine: (Moderate) Concomitant administration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent.
Lasmiditan: (Moderate) Monitor for excessive sedation and somnolence during coadministration of lasmiditan and benzodiazepines. Concurrent use may result in additive CNS depression.
Lemborexant: (Moderate) Monitor for excessive sedation and somnolence during use of lemborexant with benzodiazepines. Dosage adjustments may be necessary when administered together because of potentially additive CNS effects. Use of more than 2 hypnotics should be avoided due to the additive CNS depressant and complex sleep-related behaviors that may occur. While anxiolytic medications may be used concurrently with lemborexant, a reduction in dose of one or both agents may be needed. The risk of next-day impairment, including impaired driving, is increased if lemborexant is taken with other CNS depressants.
Levocetirizine: (Moderate) Concurrent use of cetirizine/levocetirizine with benzodiazepines should generally be avoided. Coadministration may increase the risk of CNS depressant-related side effects. If concurrent use is necessary, monitor for excessive sedation and somnolence.
Levodopa: (Moderate) Advise patients of possible additive risk for somnolence during concomitant use of levodopa with sedating agents, such as benzodiazepines. Concomitant use may also increase the risk for sudden sleep onset during activities of daily living, such as driving or eating; this adverse effect may occur without any warning signs. Patients should consider avoiding activities such as driving or operating machinery until the effects of this combination are known.
Levomilnacipran: (Moderate) Concurrent use of many CNS active drugs, including benzodiazepines, with levomilnacipran has not been evaluated by the manufacturer. Therefore, caution is advisable when combining anxiolytics, sedatives, and hypnotics or other psychoactive medications with levomilnacipran.
Levonorgestrel; Ethinyl Estradiol: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Levonorgestrel; Ethinyl Estradiol; Ferrous Bisglycinate: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Levonorgestrel; Ethinyl Estradiol; Ferrous Fumarate: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Levorphanol: (Major) Concomitant use of opiate agonists with benzodiazepines may

cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If levorphanol is initiated in a patient taking a benzodiazepine, reduce the initial dose of levorphanol by approximately 50% or more. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Lofexidine: (Moderate) Monitor for excessive hypotension and sedation during coadministration of lofexidine and benzodiazepines. Lofexidine can potentiate the effects of CNS depressants such as benzodiazepines.
Loxapine: (Moderate) The combination of loxapine and lorazepam has been associated with acute respiratory depression, stupor, and/or hypotension in several patients. Lorazepam, and possibly other benzodiazepines, should be used cautiously in patients receiving loxapine.
Lumateperone: (Moderate) Monitor for excessive sedation and somnolence during coadministration of lumateperone and benzodiazepines. Concurrent use may result in additive CNS depression.
Lurasidone: (Moderate) Due to the CNS effects of lurasidone, caution should be used when lurasidone is given in combination with other centrally acting medications such as anxiolytics, sedatives, and hypnotics, including benzodiazepines. In one study, co-administration of lurasidone and midazolam increased the Cmax and AUC of midazolam by about 21% and 44%, respectively, compared to midazolam alone; however, dosage adjustment of midazolam based upon pharmacokinetic parameters is not required during concurrent use of lurasidone.
Maprotiline: (Moderate) Benzodiazepines or other CNS depressants should be combined cautiously with maprotiline because they could cause additive depressant effects and possible respiratory depression or hypotension. The combination of benzodiazepines and maprotiline is commonly used clinically and is considered to be safe as long as patients are monitored for excessive adverse effects from either agent. Maprotiline may lower the seizure threshold, so when benzodiazepines are used for anticonvulsant effects the patient should be monitored for desired clinical outcomes.
Meclizine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Mefloquine: (Moderate) Coadministration of mefloquine and anticonvulsants may result in lower than expected anticonvulsant concentrations and loss of seizure control. Monitoring of the anticonvulsant serum concentration is recommended. Dosage adjustments may be required during and after therapy with mefloquine.
Melatonin: (Major) Use caution when combining melatonin with the benzodiazepines;

when the benzodiazepine is used for sleep, co-use of melatonin should be avoided. Use of more than 1 agent for hypnotic purposes may increase the risk for over-sedation, CNS effects, or sleep-related behaviors. Be alert for unusual changes in moods or behaviors. Use caution when combining melatonin with benzodiazepines for other uses. Patients reporting unusual sleep-related behaviors should likely discontinue melatonin use. In animal studies, melatonin has been shown to increase benzodiazepine binding to receptor sites. In one case report, a benzodiazepine-dependent woman with an 11 year history of insomnia weaned and discontinued her benzodiazepine prescription within a few days without rebound insomnia or apparent benzodiazepine withdrawal when melatonin was given. In another case report, the ingestion of excessive melatonin along with normal doses of chlordiazepoxide and an antidepressant resulted in lethargy and short-term amnestic responses. Both cases suggest additive pharmacodynamic effects. In a clinical trial, there was clear evidence for a transitory pharmacodynamic interaction between melatonin and another hypnotic agent one hour following co-dosing. Concomitant administration resulted in increased impairment of attention, memory and coordination compared to the hypnotic agent alone.
Meperidine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Meprobamate: (Moderate) Concomitant administration of benzodiazepines with meprobamate can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed.
Metaxalone: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.
Methadone: (Major) Concurrent use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective dose and minimum duration possible. If methadone is initiated for pain in an opioid-naive patient taking a benzodiazepine, use an initial methadone dose of 2.5 mg

PO every 12 hours. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial benzodiazepine dose and titrate to response. In patients treated with methadone for opioid use disorder, cessation of benzodiazepines or other CNS depressants is preferred in most cases. Consider alternatives to benzodiazepines for conditions such as anxiety or insomnia during methadone maintenance treatment. Educate patients about the risks and symptoms of respiratory depression and sedation.
Methocarbamol: (Moderate) Concurrent use of benzodiazepines and other CNS active medications including skeletal muscle relaxants, can potentiate the CNS effects of either agent. Lower doses of one or both agents may be required. The severity of this interaction may be increased when additional CNS depressants are given.
Methohexital: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use.
Methscopolamine: (Moderate) CNS depression can be increased when methscopolamine is combined with other CNS depressants such as any anxiolytics, sedatives, and hypnotics.
Methyldopa: (Moderate) Methyldopa is associated with sedative effects. Methyldopa can potentiate the effects of CNS depressants such as barbiturates, benzodiazepines, opiate agonists, or phenothiazines when administered concomitantly.
Metoclopramide: (Minor) Combined use of metoclopramide and other CNS depressants, such as anxiolytics, sedatives, and hypnotics, can increase possible sedation.
metyraPONE: (Moderate) Metyrapone may cause dizziness and/or drowsiness. Other drugs that may also cause drowsiness, such as benzodiazepines, should be used with caution. Additive drowsiness and/or dizziness is possible.
metyroSINE: (Moderate) The concomitant administration of metyrosine with benzodiazepines can result in additive sedative effects.
Milnacipran: (Moderate) Concurrent use of many CNS-active drugs with milnacipran or levomilnacipran has not been evaluated by the manufacturer. Therefore, caution is advisable when combining anxiolytics, sedatives, and hypnotics or other psychoactive medications with these medications.
Minocycline: (Minor) Injectable minocycline contains magnesium sulfate heptahydrate. Because of the CNS-depressant effects of magnesium sulfate, additive central-depressant effects can occur following concurrent administration with CNS depressants such as benzodiazepines. Caution should be exercised when using these agents concurrently.
Mirtazapine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of benzodiazepines and mirtazapine due to the risk for additive CNS depression.
Molindone: (Moderate) Consistent with the pharmacology of molindone, additive effects may occur with other CNS active drugs such as anticonvulsants. In addition, seizures

have been reported during the use of molindone, which is of particular significance in patients with a seizure disorder receiving anticonvulsants. Adequate dosages of anticonvulsants should be continued when molindone is added; patients should be monitored for clinical evidence of loss of seizure control or the need for dosage adjustments of either molindone or the anticonvulsant.
Monoamine oxidase inhibitors: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of benzodiazepines and monoamine oxidase inhibitors (MAOIs) due to the risk for additive CNS depression.
Morphine: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If morphine is initiated in a patient taking a benzodiazepine, reduce initial dosages and titrate to clinical response. For extended-release tablets, start with morphine 15 mg PO every 12 hours, and for extended-release capsules, start with 30 mg PO every 24 hours or less. Use an initial morphine; naltrexone dose of 20 mg/0.8 mg PO every 24 hours. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Nalbuphine: (Major) Concomitant use of mixed opiate agonists/antagonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of mixed opiate agonists/antagonists with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If a mixed opiate agonist/antagonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the mixed opiate agonist/antagonist and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking a mixed opiate agonist/antagonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Nitroglycerin: (Minor) Nitroglycerin can cause hypotension. This action may be additive with other agents that can cause hypotension such as benzodiazepines. Patients should be monitored more closely for hypotension if nitroglycerin is used concurrently with benzodiazepines.
Norethindrone Acetate; Ethinyl Estradiol; Ferrous fumarate: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.

Norethindrone; Ethinyl Estradiol: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Norethindrone; Ethinyl Estradiol; Ferrous fumarate: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Norgestimate; Ethinyl Estradiol: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Nortriptyline: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
OLANZapine: (Major) Concurrent use of intramuscular olanzapine and parenteral benzodiazepines is not recommended due to the potential for adverse effects from the combination including excess sedation and/or cardiorespiratory depression. Although oral formulations of olanzapine and benzodiazepines may be used together, additive effects on respiratory depression and/or CNS depression are possible. Drugs that can cause CNS depression, if used concomitantly with olanzapine, can increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, dizziness, and orthostatic hypotension. Besides ethanol, clinicians should use other anxiolytics, sedatives, and hypnotics cautiously with olanzapine.
OLANZapine; FLUoxetine: (Major) Concurrent use of intramuscular olanzapine and parenteral benzodiazepines is not recommended due to the potential for adverse effects from the combination including excess sedation and/or cardiorespiratory depression. Although oral formulations of olanzapine and benzodiazepines may be used together, additive effects on respiratory depression and/or CNS depression are possible. Drugs that can cause CNS depression, if used concomitantly with olanzapine, can increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, dizziness, and orthostatic hypotension. Besides ethanol, clinicians should use other anxiolytics, sedatives, and hypnotics cautiously with olanzapine.
OLANZapine; Samidorphan: (Major) Concurrent use of intramuscular olanzapine and parenteral benzodiazepines is not recommended due to the potential for adverse effects from the combination including excess sedation and/or cardiorespiratory depression. Although oral formulations of olanzapine and benzodiazepines may be used together, additive effects on respiratory depression and/or CNS depression are possible. Drugs that can cause CNS depression, if used concomitantly with olanzapine, can increase both the frequency and the intensity of adverse effects such as drowsiness,

sedation, dizziness, and orthostatic hypotension. Besides ethanol, clinicians should use other anxiolytics, sedatives, and hypnotics cautiously with olanzapine.
Oliceridine: (Major) Concomitant use of oliceridine with lorazepam may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opioid pain medication with lorazepam to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect.
Opicapone: (Major) Concomitant administration of benzodiazepines with other drugs have CNS depressant properties, including COMT inhibitors, can potentiate the CNS effects of either agent. COMT inhibitors have also been associated with sudden sleep onset during activities of daily living such as driving, which has resulted in accidents in some cases. Prescribers should re-assess patients for drowsiness or sleepiness regularly throughout treatment, especially since events may occur well after the start of treatment. Patients should be advised to avoid driving or other tasks requiring mental alertness until they know how the combination affects them.
Orphenadrine: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who take benzodiazepines with another CNS depressant for symptoms of excess sedation.
oxyCODONE: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If oxycodone is initiated in a patient taking a benzodiazepine, reduce dosages and titrate to clinical response. For acetaminophen; oxycodone extended-release tablets, start with 1 tablet PO every 12 hours, and for other oxycodone products, use an initial dose of oxycodone at 1/3 to 1/2 the usual dosage. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
oxyMORphone: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If oxymorphone is initiated in a patient taking a benzodiazepine, use an initial dose of oxymorphone at 1/3 to 1/2 the usual dosage and titrate to clinical response. If the extended-release oxymorphone tablets are used concurrently with a CNS depressant, use an initial dosage of 5 mg PO every 12 hours. If a benzodiazepine is

prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Paliperidone: (Moderate) Drugs that can cause CNS depression, such as benzodiazepines, can increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, and dizziness when coadministered with paliperidone. Monitor for signs and symptoms of CNS depression and advise patients to avoid driving or engaging in other activities requiring mental alertness until they know how this combination affects them.
Papaverine: (Moderate) Concurrent use of papaverine with potent CNS depressants such as benzodiazepines could lead to enhanced sedation.
Pentazocine; Naloxone: (Major) Concomitant use of mixed opiate agonists/antagonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of mixed opiate agonists/antagonists with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If a mixed opiate agonist/antagonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the mixed opiate agonist/antagonist and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking a mixed opiate agonist/antagonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
PENTobarbital: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use.
Perampanel: (Moderate) Patients taking benzodiazepines with perampanel may experience increased CNS depression. Monitor patients for adverse effects; dose adjustment of either drug may be necessary. Use of midazolam in healthy subjects who received perampanel 6 mg once daily for 20 days decreased the AUC and Cmax of midazolam by 13% and 15%, respectively, possibly due to weak induction of CYP3A4 by perampanel; the specific clinical significance of this interaction is unknown.
Perphenazine: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Perphenazine; Amitriptyline: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression. (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.

Phenelzine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of benzodiazepines and monoamine oxidase inhibitors (MAOIs) due to the risk for additive CNS depression.
PHENobarbital: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use.
PHENobarbital; Hyoscyamine; Atropine; Scopolamine: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use. (Moderate) Scopolamine may cause dizziness and drowsiness. Concurrent use of scopolamine and CNS depressants can adversely increase the risk of CNS depression.
Phenothiazines: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Phentermine; Topiramate: (Moderate) Topiramate has the potential to cause CNS depression as well as other cognitive and/or neuropsychiatric adverse reactions. The CNS depressant effects of topiramate can be potentiated pharmacodynamically by concurrent use of CNS depressant agents such as the benzodiazepines.
Phenylephrine: (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Pimozide: (Moderate) Due to the effects of pimozide on cognition, it should be used cautiously with other CNS depressants including benzodiazepines.
Pramipexole: (Major) Concomitant administration of benzodiazepines with CNS-depressant drugs, including pramipexole, can potentiate the CNS effects.
Prasterone, Dehydroepiandrosterone, DHEA (Dietary Supplements): (Major) Prasterone, dehydroepiandrosterone, DHEA may inhibit the metabolism of benzodiazepines (e.g., alprazolam, estazolam, midazolam) which undergo CYP3A4-mediated metabolism. In one study of elderly volunteers, half of the patients received DHEA 200 mg/day PO for 2 weeks, followed by a single dose of triazolam 0.25 mg. Triazolam clearance was reduced by close to 30% in the DHEA-pretreated patients vs. the control group; however, the effect of DHEA on CYP3A4 metabolism appeared to vary widely among subjects. While more study is needed, benzodiazepine-induced CNS sedation and other adverse effects might be increased in some individuals if DHEA is co-administered.
Prasterone, Dehydroepiandrosterone, DHEA (FDA-approved): (Major) Prasterone, dehydroepiandrosterone, DHEA may inhibit the metabolism of benzodiazepines (e.g., alprazolam, estazolam, midazolam) which undergo CYP3A4-mediated metabolism. In one study of elderly volunteers, half of the patients received DHEA 200 mg/day PO for 2 weeks, followed by a single dose of triazolam 0.25 mg. Triazolam clearance was reduced by close to 30% in the DHEA-pretreated patients vs. the control group; however, the effect of DHEA on CYP3A4 metabolism appeared to vary widely among subjects. While more study is needed, benzodiazepine-induced CNS sedation and other adverse effects

might be increased in some individuals if DHEA is co-administered.
Pregabalin: (Major) Concomitant use of benzodiazepines with pregabalin may cause excessive sedation, somnolence, and respiratory depression. If concurrent use is necessary, initiate pregabalin at the lowest recommended dose and monitor patients for symptoms of respiratory depression and sedation. Educate patients about the risks and symptoms of excessive CNS depression and respiratory depression.
Primidone: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use.
Probenecid: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and probenecid is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Coadministration of lorazepam with probenecid may cause a more rapid onset or prolonged effect of lorazepam due to increased half-life and decreased total clearance. Lorazepam is an UGT substrate and probenecid is an UGT inhibitor.
Probenecid; Colchicine: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and probenecid is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Coadministration of lorazepam with probenecid may cause a more rapid onset or prolonged effect of lorazepam due to increased half-life and decreased total clearance. Lorazepam is an UGT substrate and probenecid is an UGT inhibitor.
Procarbazine: (Minor) CNS depressants benzodiazepines can potentiate the CNS depression caused by procarbazine therapy, so these drugs should be used together cautiously.
Prochlorperazine: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Promethazine: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Promethazine; Dextromethorphan: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Promethazine; Phenylephrine: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression. (Moderate) The therapeutic effect of phenylephrine may be decreased in patients receiving benzodiazepines. Monitor patients for decreased pressor effect if these agents are administered concomitantly.
Propofol: (Moderate) Concomitant administration can potentiate the CNS effects (e.g.,

increased sedation or respiratory depression) of either agent.
Protriptyline: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
Pseudoephedrine; Triprolidine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Pyrimethamine: (Moderate) Mild hepatotoxicity has been reported when pyrimethamine was coadministered with lorazepam.
QUEtiapine: (Moderate) Monitor for excessive sedation and somnolence during coadministration of alprazolam and quetiapine. Concurrent use may result in additive CNS depression. Quetiapine decreases lorazepam clearance by about 20%.
Ramelteon: (Moderate) Ramelteon is a sleep-promoting agent; therefore, additive pharmacodynamic effects are possible when combining ramelteon with benzodiazepines or other miscellaneous anxiolytics, sedatives, and hypnotics. Pharmacokinetic interactions have been observed with the use of zolpidem. Use of ramelteon 8 mg/day for 11 days and a single dose of zolpidem 10 mg resulted in an increase in the median Tmax of zolpidem of about 20 minutes; exposure to zolpidem was unchanged. Ramelteon use with hypnotics of any kind is considered duplicative therapy and these drugs are generally not co-administered.
Rasagiline: (Moderate) The CNS-depressant effects of MAOIs can be potentiated with concomitant administration of other drugs known to cause CNS depression including buprenorphine, butorphanol, dronabinol, THC, nabilone, nalbuphine, and anxiolytics, sedatives, and hypnotics. Use these drugs cautiously with MAOIs; warn patients to not drive or perform other hazardous activities until they know how a particular drug combination affects them. In some cases, the dosages of the CNS depressants may need to be reduced.
Remifentanil: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Benzodiazepine doses may need to be reduced up to 75% during coadministration with remifentanil. Educate patients about the risks and symptoms of respiratory depression and sedation.
Remimazolam: (Major) The sedative effect of remimazolam can be accentuated by

lorazepam. Titrate the dose of remimazolam to the desired clinical response and continuously monitor sedated patients for hypotension, airway obstruction, hypoventilation, apnea, and oxygen desaturation.
risperiDONE: (Moderate) Due to the primary CNS effects of risperidone, caution should be used when risperidone is given in combination with other centrally acting medications including anxiolytics, sedatives, and hypnotics.
rOPINIRole: (Moderate) Concomitant use of ropinirole with other CNS depressants can potentiate the sedation effects of ropinirole.
Rotigotine: (Major) Concomitant use of rotigotine with other CNS depressants, such as benzodiazepines, can potentiate the sedative effects of rotigotine.
Safinamide: (Moderate) Dopaminergic medications, including safinamide, may cause a sudden onset of somnolence which sometimes has resulted in motor vehicle accidents. Patients may not perceive warning signs, such as excessive drowsiness, or they may report feeling alert immediately prior to the event. Because of possible additive effects, advise patients about the potential for increased somnolence during concurrent use of safinamide with other sedating medications, such as benzodiazepines.
Scopolamine: (Moderate) Scopolamine may cause dizziness and drowsiness. Concurrent use of scopolamine and CNS depressants can adversely increase the risk of CNS depression.
Secobarbital: (Moderate) Additive CNS and/or respiratory depression may occur with concurrent use.
Sedating H1-blockers: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Segesterone Acetate; Ethinyl Estradiol: (Minor) Ethinyl estradiol may enhance the metabolism of lorazepam. It appears glucuronide conjugation of lorazepam is increased in the presence of combined hormonal oral contraceptives; the clinical significance of this interaction is not determined.
Selegiline: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of benzodiazepines and selegiline due to the risk for additive CNS depression.
Sevoflurane: (Moderate) Concomitant administration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent.
Sincalide: (Moderate) Sincalide-induced gallbladder ejection fraction may be affected by benzodiazepines. False study results are possible in patients with drug-induced hyper- or hypo-responsiveness; thorough patient history is important in the interpretation of procedure results.
Skeletal Muscle Relaxants: (Moderate) Concomitant use of skeletal muscle relaxants with benzodiazepines can result in additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given. Monitor patients who

take benzodiazepines with another CNS depressant for symptoms of excess sedation.
Sodium Oxybate: (Contraindicated) Sodium oxybate should not be used in combination with CNS depressant anxiolytics, sedatives, and hypnotics or other sedative CNS depressant drugs. Specifically, sodium oxybate use is contraindicated in patients being treated with sedative hypnotic drugs. Sodium oxybate (GHB) has the potential to impair cognitive and motor skills. For example, the concomitant use of barbiturates and benzodiazepines increases sleep duration and may contribute to rapid onset, pronounced CNS depression, respiratory depression, or coma when combined with sodium oxybate.
SORAfenib: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and sorafenib is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Lorazepam is an UGT substrate and sorafenib is an UGT inhibitor.
Stiripentol: (Moderate) Monitor for excessive sedation and somnolence during coadministration of stiripentol and lorazepam. CNS depressants can potentiate the effects of stiripentol.
SUFentanil: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Sulopenem Etzadroxil; Probenecid: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and probenecid is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily titrated. Coadministration of lorazepam with probenecid may cause a more rapid onset or prolonged effect of lorazepam due to increased half-life and decreased total clearance. Lorazepam is an UGT substrate and probenecid is an UGT inhibitor.
Suvorexant: (Moderate) CNS depressant drugs may have cumulative effects when administered concurrently and they should be used cautiously with suvorexant. A reduction in dose of the CNS depressant may be needed in some cases. These agents include the benzodiazepines.
Tapentadol: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use

of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If tapentadol is initiated in a patient taking a benzodiazepine, a reduced initial dosage of tapentadol is recommended. If the extended-release tapentadol tablets are used concurrently with a benzodiazepine, use an initial tapentadol dose of 50 mg PO every 12 hours. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Teduglutide: (Moderate) Altered mental status has been observed in patients taking teduglutide and benzodiazepines in the adult clinical studies for teduglutide. Careful monitoring and possible dose adjustment of the benzodiazepine agent may be required. Teduglutide has direct effects on the gut that may increase benzodiazepine exposure by improving oral absorption.
Tetrabenazine: (Moderate) Concurrent use of tetrabenazine and drugs that can cause CNS depression, such as benzodiazepines, can increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, dizziness, and orthostatic hypotension.
Thalidomide: (Major) The use of benzodiazepine anxiolytics, sedatives, or hypnotics with thalidomide may cause an additive sedative effect and should be avoided. Thalidomide frequently causes drowsiness and somnolence. Dose reductions may be required. Patients should be instructed to avoid situations where drowsiness may be a problem and not to take other medications that may cause drowsiness without adequate medical advice. Advise patients as to the possible impairment of mental and/or physical abilities required for the performance of hazardous tasks, such as driving a car or operating other complex or dangerous machinery.
Theophylline, Aminophylline: (Minor) Aminophylline or Theophylline have been reported to counteract the pharmacodynamic effects of diazepam and possibly other benzodiazepines. The clinical significance of this interaction is not certain. A proposed mechanism is competitive binding of these methylxanthines to adenosine receptors in the brain. If such therapy is initiated or discontinued, monitor the clinical response to the benzodiazepine.
Thioridazine: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Thiothixene: (Moderate) Thiothixene can potentiate the CNS-depressant action of other drugs such as benzodiazepines. Caution should be exercised during simultaneous use of these agents due to potential excessive CNS effects or additive hypotension.
tiaGABine: (Moderate) Because of the possible additive effects of drugs that depress the

central nervous system, benzodiazepines should be used with caution in patients receiving tiagabine.
tiZANidine: (Moderate) Concurrent use of tizanidine and CNS depressants like the benzodiazepines can cause additive CNS depression. The severity of this interaction may be increased when additional CNS depressants are given.
Tolcapone: (Major) Concomitant administration of benzodiazepines with other drugs have CNS depressant properties, including COMT inhibitors, can potentiate the CNS effects of either agent. COMT inhibitors have also been associated with sudden sleep onset during activities of daily living such as driving, which has resulted in accidents in some cases. Prescribers should re-assess patients for drowsiness or sleepiness regularly throughout treatment, especially since events may occur well after the start of treatment. Patients should be advised to avoid driving or other tasks requiring mental alertness until they know how the combination affects them.
Topiramate: (Moderate) Topiramate has the potential to cause CNS depression as well as other cognitive and/or neuropsychiatric adverse reactions. The CNS depressant effects of topiramate can be potentiated pharmacodynamically by concurrent use of CNS depressant agents such as the benzodiazepines.
traMADol: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Tramadol; Acetaminophen: (Major) Concomitant use of opiate agonists with benzodiazepines may cause respiratory depression, hypotension, profound sedation, and death. Limit the use of opiate pain medications with benzodiazepines to only patients for whom alternative treatment options are inadequate. If concurrent use is necessary, use the lowest effective doses and minimum treatment durations needed to achieve the desired clinical effect. If an opiate agonist is initiated in a patient taking a benzodiazepine, use a lower initial dose of the opiate and titrate to clinical response. If a benzodiazepine is prescribed for an indication other than epilepsy in a patient taking an opiate agonist, use a lower initial dose of the benzodiazepine and titrate to clinical response. Educate patients about the risks and symptoms of respiratory depression and sedation.
Tranylcypromine: (Moderate) Monitor for unusual drowsiness and sedation during coadministration of benzodiazepines and monoamine oxidase inhibitors (MAOIs) due to

the risk for additive CNS depression.
traZODone: (Major) Monitor for excessive sedation and somnolence during coadministration of trazodone and benzodiazepines. Concurrent use may result in additive CNS depression.
Tricyclic antidepressants: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
Trifluoperazine: (Major) Limit dosage and duration of benzodiazepines during concomitant phenothiazine use and monitor for unusual drowsiness and sedation due to the risk for additive CNS depression.
Trihexyphenidyl: (Moderate) CNS depressants, such as anxiolytics, sedatives, and hypnotics, can increase the sedative effects of trihexyphenidyl.
Trimethobenzamide: (Moderate) The concurrent use of trimethobenzamide with other medications that cause CNS depression, like the benzodiazepines, may potentiate the effects of either trimethobenzamide or the benzodiazepine.
Trimipramine: (Major) Limit dosage and duration of benzodiazepines during concomitant use with tricyclic antidepressants, and monitor patients closely for respiratory depression and sedation. Additive CNS depression may occur.
Triprolidine: (Moderate) Coadministration can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. Use caution with this combination.
Valerian, Valeriana officinalis: (Major) Any substances that act on the CNS, including psychoactive drugs and drugs used as anesthetic adjuvants (e.g., barbiturates, benzodiazepines), may theoretically interact with valerian, Valeriana officinalis. The valerian derivative, dihydrovaltrate, binds at barbiturate binding sites; valerenic acid has been shown to inhibit enzyme-induced breakdown of GABA in the brain; the non-volatile monoterpenes (valepotriates) have sedative activity. These interactions are probably pharmacodynamic in nature. There is a possibility of interaction with valerian at normal prescription dosages of anxiolytics, sedatives, and hypnotics (including barbiturates and benzodiazepines). Patients who are taking barbiturates or other sedative/hypnotic drugs should avoid concomitant administration of valerian. Patients taking medications such as tricyclic antidepressants, lithium, MAOIs, skeletal muscle relaxants, SSRIs and serotonin norepinephrine reuptake inhibitors (e.g., duloxetine, venlafaxine) should discuss the use of herbal supplements with their health care professional prior to consuming valerian; combinations should be approached with caution in the absence of clinical data. Patients should not abruptly stop taking their prescribed psychoactive medications.
Valproic Acid, Divalproex Sodium: (Moderate) Monitor for an increase in lorazepam-related adverse reactions and consider reducing the dose of lorazepam if concomitant use of lorazepam and valproic acid is necessary. Avoid lorazepam extended-release capsules and utilize lorazepam immediate-release dosage forms that can be easily

titrated. Coadminstration of lorazepam with valproic acid causes increased plasma concentrations and reduced clearance of lorazepam. Lorazepam is an UGT substrate and valproic acid is an UGT inhibitor.
Vancomycin: (Moderate) The concurrent administration of vancomycin and anesthetics has been associated with erythema, histamine-like flushing, and anaphylactoid reactions.
Vigabatrin: (Moderate) Vigabatrin may cause somnolence and fatigue. Drugs that can cause CNS depression, if used concomitantly with vigabatrin, may increase both the frequency and the intensity of adverse effects such as drowsiness, sedation, and dizziness. Caution should be used when vigabatrin is given in combination with benzodiazepines.
Vilazodone: (Moderate) Due to the CNS effects of vilazodone, caution should be used when vilazodone is given in combination with other centrally acting medications such as the benzodiazepines.
Zaleplon: (Major) Monitor for excessive sedation and somnolence during coadministration of zaleplon and benzodiazepines. Concurrent use may result in additive CNS depression. If used together, a reduction in the dose of one or both drugs may be needed.
Ziprasidone: (Moderate) Ziprasidone has the potential to impair cognitive and motor skills. Additive CNS depressant effects are possible when ziprasidone is used concurrently with any CNS depressant.
Zolpidem: (Major) Concomitant administration of benzodiazepines with zolpidem can potentiate the CNS effects (e.g., increased sedation or respiratory depression) of either agent. If used together, a reduction in the dose of one or both drugs may be needed. For Intermezzo brand of sublingual zolpidem tablets, reduce the dose to 1.75 mg/night. Concurrent use of zolpidem with other sedative-hypnotics, including other zolpidem products, at bedtime or the middle of the night is not recommended. In addition, sleep-related behaviors, such as sleep-driving, are more likely to occur during concurrent use of zolpidem and other CNS depressants than with zolpidem alone.
Zuranolone: (Major) Avoid the use of multiple sedating agents due to the risk for additive CNS depression. If use is necessary, consider a downward dosage adjustment of either or both medications, especially in patients with additional risk factors for sedation-related harm.

Adverse Reaction

agitation, asthenia, ataxia, coma, confusion, depression, dizziness, drowsiness, dysarthria, euphoria, fatigue, hallucinations, headache, muscle paralysis, neuroleptic malignant syndrome-like symptoms, suicidal ideation, tremor, vertigo, weakness

Most of the CNS-related adverse effects associated with lorazepam therapy are dose-dependent. The most frequent adverse effects in adult patients with anxiety treated with oral lorazepam included drowsiness (15.9%), dizziness (6.9%), weakness (4.2%), and unsteadiness (3.4%); the incidence of drowsiness and unsteadiness increased with increasing adult age. Because lorazepam can cause drowsiness and a decreased level of consciousness, there is a higher risk of falls, particularly in the older adult, with the potential for subsequent severe injuries. Other CNS events occurring after oral lorazepam therapy included confusion, disorientation, depression, the unmasking of depression, disinhibition, suicidal ideation/attempt, ataxia, asthenia, extrapyramidal symptoms, tremor, vertigo, fatigue, euphoria, headache, coma, dysarthria, and autonomic manifestations. CNS adverse events occurring in clinical trials of parenteral lorazepam for the treatment of status epilepticus included somnolence/drowsiness (1.5% to 3.5%), headache (0.1% to 1.2%), coma (0.1% to 1.2%), and stupor (0.1% to 1.2%). Agitation, asthenia, ataxia, brain edema, confusion, hallucinations, myoclonus, abnormal thinking, and tremor were also reported in 0.1% to 1% of patients. Myoclonus has particularly been associated with lorazepam injection use in very low birth weight neonates. Confusion and depression occurred in approximately 1.3% of patients given parenteral lorazepam as a pre-anesthetic. Muscle paralysis, neuroleptic malignant syndrome-like symptoms, nervousness, acute brain syndrome, coma, and brain edema have been reported during postmarketing use of parenteral lorazepam; causality to the drug has not been established. CNS depression may be severe and result in profound sedation, along with depressed respiration, and may occasionally result in death. Fatalities have occurred when benzodiazepines are coadministered with opioid medications and/or alcohol.

amnesia, memory impairment

Anterograde amnesia and memory impairment can occur with therapeutic doses of benzodiazepines, such as lorazepam. Amnesia and memory impairment are clinically desirable during critical care sedation, perioperative periods, or prior to or during certain medical procedures. The risk for developing anterograde amnesia increases at higher doses and with concurrent ingestion of alcohol.

anxiety, delirium, hostility, hyperactivity, hyperreflexia, insomnia, irritability, mania, nightmares, psychosis, restlessness

Occasionally, paradoxical CNS stimulation can occur, particularly in patients with psychiatric disorders, geriatric adults, and children. Symptoms of paradoxical CNS stimulation due to benzodiazepines include irritability, hostility, psychosis, aggression, rage, nightmares, talkativeness, excitement, restlessness, hyperactivity, mania,

tremulousness, sleep disturbances, insomnia, increased muscle spasticity, anxiety, and hyperreflexia. CNS symptoms, such as restlessness, delirium, crying, and sobbing occurred in approximately 1.3% of patients given parenteral lorazepam as a pre-anesthetic. Paradoxical excitation was observed in 10% to 30% of children less than 8 years who were treated with lorazepam injection for status epilepticus. Lorazepam should be discontinued if any of the signs of paradoxical CNS excitement occur.

apnea, hyperventilation, hypoventilation, hypoxia, pneumothorax, pulmonary edema, pulmonary hypertension, respiratory arrest, respiratory depression

Respiratory depression can occur with benzodiazepine therapy, particularly with misuse, coadministration with other CNS depressants, or overdosage. Concomitant use of benzodiazepines and opioids or alcohol may result in profound sedation, respiratory depression, coma, and death. Apnea has been reported following parenteral administration of benzodiazepines, such as lorazepam, to the elderly, severely ill patients, or those patients with compromised respiratory function. Respiratory adverse events occurring in clinical trials of parenteral lorazepam for the treatment of status epilepticus included respiratory failure (1.5% to 2.4%), apnea (at least 1%), hyperventilation (less than 1%), hypoventilation (0.1% to 1.2%), and respiratory disorder (0.1% to 1.2%). Airway obstruction was reported rarely in patients receiving parenteral lorazepam as a pre-anesthetic. Respiratory effects reported during oral lorazepam treatment have included respiratory depression, apnea, worsening of sleep apnea, and worsening of obstructive pulmonary disease. Respiratory events such as pulmonary edema, lung hemorrhage, pneumothorax, respiratory arrest, and pulmonary hypertension have been reported with parenteral lorazepam postmarketing; however, a causal relationship has not been established. Hypoventilation, airway obstruction, and apnea can lead to hypoxia and/or cardiovascular or respiratory arrest unless effective countermeasures are taken immediately.

myoclonia, seizures

Seizures occurred in less than 1% of patients receiving parenteral lorazepam for the treatment of status epilepticus in clinical trials. Seizures and myoclonia/myoclonus have been reported after administration of parenteral lorazepam, especially in very low birth weight infants. Tonic-clonic seizures have occurred after administration of parenteral lorazepam to patients with atypical petit mal status epilepticus but the incidence is thought to be rare. Seizures have also been reported with oral lorazepam. Although uncommon, seizures may occur following the discontinuation of benzodiazepine therapy.

constipation, diarrhea, elevated hepatic enzymes, GI bleeding, hyperbilirubinemia, hypersalivation, jaundice, nausea, vomiting

Gastrointestinal (GI) adverse effects occurring in less than 1% of patients receiving parenteral lorazepam for status epilepticus included nausea, vomiting, elevated hepatic enzymes, and hypersalivation. Nausea, jaundice, hyperbilirubinemia, constipation, change in appetite, and elevated hepatic enzymes have been reported with oral lorazepam. GI bleeding and liver damage have been reported in postmarketing reports of parenteral lorazepam. Elevations of LDH have developed in patients receiving lorazepam and liver function tests are recommended for patients on long-term therapy. Because lorazepam oral concentrate contains polyethylene glycol and propylene glycol, high doses may result in osmotic diarrhea. If drug-induced diarrhea occurs, discontinue the oral concentrate and substitute the dose with lorazepam tablets. [

erythema, injection site reaction, tissue necrosis

Injection site reaction, including erythema and pain, has been reported after administration of parenteral lorazepam and ranges in frequency depending on the route of administration. After intramuscular injection, erythema occurred immediately in approximately 2% of patients and was still present after 24 hours in 0.8%. Erythema did not occur immediately in patients who received intravenous lorazepam but was present in 2.4% of patients at the 24-hour observation period. Pain and/or burning was greatest in the immediate period after intramuscular administration (17%) and was still present in a small percentage of patients (0.5% to 1.4%) 24 hours after parenteral administration. Of those who received intravenous lorazepam, 1.6% experienced pain immediately after administration and 0.5% still complained of pain 24 hours later. Inadvertent intra-arterial injection of lorazepam may produce arteriospasm resulting in gangrene (tissue necrosis and ischemia) which may require amputation.

chills, cystitis, hypothermia, infection

Infection, chills, and cystitis occurred in less than 1% of patients receiving parenteral lorazepam in clinical trials for the treatment of status epilepticus. Hypothermia has been reported during treatment with oral lorazepam. Aggravation of pheochromocytoma has been reported during postmarketing use of injectable lorazepam; however, causality to the drug has not been established.

hyponatremia, lactic acidosis, metabolic acidosis, renal tubular necrosis, SIADH

Metabolic acidosis occurred in less than 1% of patients receiving parenteral lorazepam

in clinical trials for the treatment of status epilepticus. There also have been reports of propylene glycol toxicity (e.g. lactic acidosis, hyperosmolality, hypotension) and possible polyethylene toxicity (e.g. acute renal tubular necrosis) during the administration of lorazepam injection or oral lorazepam solution at higher than recommended doses. SIADH and hyponatremia have been reported during oral lorazepam therapy; however, the frequencies are unknown and causality to the drug is not established.

impotence (erectile dysfunction), libido decrease, orgasm dysfunction, urinary incontinence

Adverse genitourinary (GU) effects reported during oral lorazepam therapy include a change in libido (unspecified), orgasm dysfunction, and impotence (erectile dysfunction). Urinary incontinence has been reported during postmarketing use of parenteral lorazepam. Adverse GU/sexual effects including menstrual irregularity, urinary incontinence, urinary retention, and libido decrease have been reported as potential side effects of other benzodiazepines.

alopecia, anaphylactoid reactions, rash

Hypersensitivity reactions, anaphylactoid reactions, dermatological symptoms, allergic skin reactions (rash, unspecified), and alopecia have been reported during lorazepam therapy. Skin rash has been reported with parenteral use in adult surgical patients.

AV block, bradycardia, cardiac arrest, heart failure, hypertension, hypotension, pericardial effusion, sinus tachycardia

Hypotension and cardiac arrest have been reported following parenteral administration of benzodiazepines, such as lorazepam, to the elderly, severely ill patients, or to patients with compromised respiratory function. Hypotension occurred in 1.5% to 2.4% of patients receiving parenteral lorazepam for the treatment of status epilepticus in clinical trials. Small and usually insignificant decreases in blood pressure or hypotension may also occur with oral lorazepam and is likely related to relief of anxiety resulting from lorazepam administration. Hypertension has occasionally been reported with parenteral lorazepam (0.1% of patients). Cardiovascular events such as ventricular arrhythmias, bradycardia, heart failure, AV block, pericardial effusion, and sinus tachycardia have been reported in postmarketing reports of parenteral lorazepam; however, a casual relationship to the drug has not been established. Hypoventilation, airway obstruction, and apnea can lead to hypoxia and/or cardiac arrest unless effective countermeasures are taken immediately. Cardiac arrest is rare but has been fatal.

blurred vision, diplopia, hearing loss

Adverse ocular effects or visual disturbances reported during treatment with oral and parenteral lorazepam include blurred vision and diplopia. Some benzodiazepines have been associated with an increased risk for acute angle closure glaucoma; however, in one epidemiological study, lorazepam was not associated with this effect. Adult patients receiving parenteral lorazepam as a pre-anesthetic infrequently reported depressed hearing (hearing loss) during clinical trials.

agranulocytosis, coagulopathy, leukopenia, pancytopenia, thrombocytopenia

Adverse hematologic effects reported with unknown frequency during treatment with lorazepam include leukopenia, thrombocytopenia, agranulocytosis, and pancytopenia. Coagulopathy has been reported during postmarketing use of injectable lorazepam; however, causality to the drug has not been established. Periodic blood counts are recommended for patients on long-term therapy.

tolerance

Tolerance to lorazepam may develop from continued therapy. Tolerance is a physiological state characterized by a reduced response to a drug after repeated administration (i.e., a higher dose of a drug is required to produce the same effect that was once obtained at a lower dose). Tolerance to the therapeutic effect of lorazepam may develop; however, little tolerance develops to the amnestic reactions and other cognitive impairments caused by benzodiazepines.

physiological dependence, psychological dependence, withdrawal

Prolonged use of benzodiazepines can produce physiological dependence, with or without psychological dependence. The risk of psychological dependence increases in those with a history of alcohol or substance abuse. Abuse and misuse of benzodiazepines commonly involve concomitant use of other medications, alcohol, and/or illicit substances, which is associated with an increased frequency of serious adverse outcomes, including respiratory depression, overdose, and death. Caution is advised when considering the use of benzodiazepines in patients with a known or suspected history of substance abuse. To discourage abuse, prescribe the smallest appropriate quantity of the benzodiazepine and provide proper disposal instructions for unused drug. Avoid or minimize concomitant use of CNS depressants or other medications associated with addiction or abuse. Clinically significant physiological dependence may occur with continued use of benzodiazepines. The risks of dependence and withdrawal increase with longer treatment duration and higher daily doses. Abrupt

discontinuation or rapid dosage reduction of benzodiazepines after continued use may precipitate acute withdrawal reactions, which can be life-threatening. Withdrawal symptoms may range from mild dysphoria and insomnia to a major withdrawal syndrome including abdominal and muscle cramps, vomiting, sweating, tremors, and seizures. In some cases, benzodiazepine users have developed a protracted withdrawal syndrome with withdrawal symptoms lasting weeks to more than 12 months. Use a gradual, patient-specific taper to lower the risk of withdrawal reactions when reducing the dose or discontinuing therapy; if significant withdrawal symptoms occur, the rate of reduction may need to be slowed or paused. Adjunctive medications may also be considered to help mitigate withdrawal effects.

neonatal abstinence syndrome, neonatal respiratory depression

Benzodiazepines cross the placenta and may produce sedation, hypotonia, lethargy, and neonatal respiratory depression at birth. Monitor neonates exposed during pregnancy or labor for signs of sedation, respiratory depression, hypotonia, and feeding problems. Neonatal withdrawal or a neonatal abstinence syndrome has also been reported following the ingestion of therapeutic doses of benzodiazepines during the last weeks of pregnancy. Clinical manifestations of neonatal withdrawal may include hyperreflexia, irritability, restlessness, tremors, inconsolable crying, and feeding difficulties. These complications can appear shortly after delivery to 3 weeks after birth and persist from hours to several months, depending on the degree of dependence and pharmacokinetic profile of the benzodiazepine. Observe neonates who are exposed to benzodiazepines in utero during the later stages of pregnancy for symptoms of withdrawal and manage appropriately. Benzodiazepines should be withdrawn cautiously and slowly, using a very gradual dosage-tapering schedule.

Description

Lorazepam is an intermediate-acting oral and parenteral benzodiazepine indicated for the treatment of anxiety, status epilepticus (parenteral), perioperative sedation or amnesia induction (parenteral), and the short-term treatment of insomnia secondary to anxiety or transient situational stress. Common off-label uses include the adjunctive treatment of chemotherapy-induced nausea/vomiting and to treat acute alcohol withdrawal. When treating anxiety disorders, benzodiazepines are generally reserved for the temporary management of anxiety symptoms as an adjunct to first-line pharmacotherapies (e.g., SSRI or SNRI) or for treatment-resistant cases. Lorazepam is often preferred over long-acting benzodiazepines (e.g., diazepam) in people with hepatic impairment or in geriatric adults because of a shorter half-life and no active metabolites. 68581

Benzodiazepines should generally be prescribed for short periods only due to the potential for dependence and cognitive impairment, and the availability of safer alternatives. The ASAM guidelines recommend re-evaluating the risks and benefits of ongoing benzodiazepine use at least every 3 months; for patients who are candidates for discontinuation, an individualized taper strategy should be used that considers dose, duration, and frequency of use. If possible, avoid coadministration of benzodiazepines and opioids due to the potential for profound sedation, respiratory depression, coma, and death; patients receiving the combination should be monitored for respiratory depression and sedation.

Mechanism Of Action

Benzodiazepines act at the level of the limbic, thalamic, and hypothalamic regions of the CNS, and can produce any level of CNS depression required including sedation, hypnosis, skeletal muscle relaxation, anticonvulsant activity, and coma. The action of these drugs is mediated through the inhibitory neurotransmitter gamma-aminobutyric acid (GABA). Central benzodiazepine receptors interact allosterically with GABA receptors, potentiating the effects of GABA and thereby increasing the inhibition of the ascending reticular activating system. Benzodiazepine activity shows the highest affinity for GABA subtype A receptor modulation compared to subtype B receptors. Benzodiazepines block the cortical and limbic arousal that occurs following stimulation of the reticular pathways. Studies in healthy volunteers show that in single high doses, lorazepam has a tranquilizing action on the central nervous system with usually no appreciable effect on the respiratory or cardiovascular systems.

Pharmacokinetics

Lorazepam is administered orally and parenterally. The drug has also been given sublingually; although, specific sublingual dosage forms are not available in the United States. Lorazepam is lipophilic; it is widely distributed and crosses the blood-brain barrier. It is approximately 85% protein-bound. In healthy adults, reported mean volume of distributions (Vd) are 1.3 L/kg following parenteral administration and 117 L following a single 3 mg dose of the extended-release capsules under fasting conditions. The Vd is smaller in neonates and slightly larger in non-neonatal pediatric patients. Lorazepam is conjugated by the liver via UDP-glucuronosyltransferase (UGT) to lorazepam glucuronide, an inactive metabolite. There is no evidence of accumulation of lorazepam with administration up to 6 months. Reported elimination half-lives are 12 hours, 14 +/- 5 hours, and 20.2 +/- 7.2 hours for immediate-release oral formulations, the parenteral formulation, and the extended-release capsules, respectively. Lorazepam clearance is significantly slower in neonates compared to adults; clearance in older children is

dependent on the specific population and varies from slightly slower to slightly faster than that of adults. Lorazepam in excreted in the urine primarily as the inactive glucuronide metabolite; lorazepam also undergoes enterohepatic recirculation. Oral mean plasma clearance (CL/F) is approximately 72 mL/minute in adults following a single 3 mg dose of the extended-release capsules.

Affected cytochrome P450 isoenzymes and drug transporters: UGT
Lorazepam is a substrate of UDP-glucuronosyltransferase (UGT). Concurrent administration of lorazepam with a UGT inhibitor may result in increased plasma concentrations, reduced clearance, and prolonged half-life of lorazepam.

Route-Specific Pharmacokinetics

- **Oral Route**

Immediate-release tablets and solution: Lorazepam is readily absorbed following an oral dose, with an absolute bioavailability of 90% reported following administration of immediate-release tablets. Subjective central nervous system effects occur within 1 to 2 hours; peak plasma concentrations occur 2 hours following administration. Mean area under concentration curve (AUCTau), Cmax, and Cmin were 765 ng x hour/mL, 41 ng/mL and 29 ng/mL, respectively, following 3 times daily administration of 1 mg tablets.

Extended-release (ER) capsules: Pharmacokinetics of the extended-release capsules are dose proportional over the dose range of 1 to 3 mg. Steady-state is usually achieved following 5 days of administration. At steady state, AUCTau, Cmax, and Cmin were 694 ng x hour/mL, 35 ng/mL and 25 ng/mL, respectively, following once daily administration of the 3 mg ER capsules. Median Tmax was 14 hours (range 7 to 24 hours) following a single 3 mg dose of the extended-release capsules. Administration of the extended-release capsules with a high-fat and high calorie meal delayed median Tmax by approximately 2 hours and did not affect overall drug exposure. Administration of the extended-release capsules by sprinkling the contents in 15 mL of applesauce did not significantly affect overall drug exposure or Tmax.

- **Intravenous Route**

Optimum anxiolytic and sedative effects occur within 15 to 20 minutes after administration; however, the onset of effect occurs more rapidly. The degree of sedation is dependent on the dose administered and the presence or absence of other medications. When used as an anticonvulsant, cessation of seizure activity may occur within 5 minutes. After IV administration of a 4 mg dose to adult patients, initial concentrations are approximately 70 ng/mL. Plasma concentrations are proportional to the dose given. Based on non-neonatal pediatric pharmacokinetic models, lorazepam 0.1 mg/kg (up to 4 mg) is expected to achieve a Cmax of 100 ng/mL; concentrations greater than 30 ng/mL are expected to be maintained for 6 to 12 hours for most

pediatric patients. After the initial dose, a second dose of 0.05 mg/kg (up to 2 mg) is expected to maintain a typical desired concentration for seizure suppression (more than 50 ng/mL) for approximately 12 hours. The duration of the sedative effect is approximately 6 to 12 hours for most patients.

- **Intramuscular Route**

Lorazepam is absorbed rapidly and completely after intramuscular injection with a bioavailability more than 90%. Optimum anxiolytic and sedative effects occur approximately 1 to 2 hours after administration, with the degree of sedation dependent on the dose administered and the presence or absence of other medications. After administration of 4 mg IM to adult patients, peak concentrations of approximately 48 ng/mL are reached within 3 hours.

- **Hepatic Impairment**

Because lorazepam undergoes conjugative metabolism as opposed to oxidative metabolism, the pharmacokinetics of the drug are less likely to be altered in the presence of hepatic dysfunction versus many other benzodiazepines. Dosages for patients with severe hepatic insufficiency should be adjusted according to patient response; lower doses may be sufficient in such patients.

- **Renal Impairment**

Renal impairment is expected to decrease the elimination of lorazepam glucuronide, but there should be no direct effect on the conversion of lorazepam to lorazepam glucuronide (i.e., inactivation of lorazepam). In 6 adults with renal impairment (CrCl 22 mL/minute), the mean volume of distribution and terminal elimination half-life were 40% and 25% higher, respectively, compared to those of healthy adults. However, the mean total clearance did not change. In 4 adults receiving chronic hemodialysis, the mean Vd and terminal elimination half-life were 75% higher compared to those of healthy adults; however, the mean total clearance was not different. During a 6-hour dialysis session, 8% of the administered dose was removed as intact lorazepam and 40% of the administered dose was removed as lorazepam glucuronide.

- **Pediatrics**

Infants, Children, and Adolescents

Infants, children, and adolescents have slightly longer half-lives and larger volumes of distribution (Vd) compared to adults. In a study of children and adolescents 2 years and older with acute lymphocytic leukemia, the mean Vd was 50% higher than that of healthy adults. Mean half-life was 30% longer in children and 2-fold greater in adolescents. In a pharmacokinetic study of 63 patients aged 5 months to 17 years, mean plasma clearance (CL) and elimination half-life of IV lorazepam (0.05 to 0.1 mg/kg/dose) was 1.2 mL/kg/minute and 16.8 hours (range 6 to 42 hours), respectively. Mean Vd was 1.48 L/kg. In general, Vd and CL decreased and half-life increased with ascending age. Reported data is as follows: for infants and children 5 months to 2 years, Vd = 1.62

L/kg, half-life = 15.8 hours, CL = 1.57 mL/kg/minute; for children 3 to 12 years, Vd = 1.5 L/kg, half-life = 16.9 hours, CL = 1.12 mL/kg/minute; for adolescents 13 to 18 years: Vd = 1.27 L/kg, half-life = 17.8 hours, CL = 0.95 mL/kg/minute.

Neonates
Neonates have significantly longer half-lives and slower clearance than other populations. After a single dose IV lorazepam (0.05 to 0.1 mg/kg) in 10 term neonates (gestational age 37 to 41 weeks) with asphyxia neonatorum presenting with seizures in the first 48 hours of life, the mean plasma clearance (CL) was 0.232 mL/kg/minute and the mean elimination half-life was 40.2 hours (range 18 to 73 hours), respectively. These values are most likely affected by altered protein binding and decreased hepatic enzyme activity in neonates. Volume of distribution (Vd) was 0.76 L/kg (range 0.14 to 1.3 L/kg); this smaller Vd compared to adults is consistent with the lower percentage of adipose tissue in the neonatal population.

- **Geriatric**

A comparison of elderly adults 66 years and older to younger adults revealed that lorazepam pharmacokinetics remain unaltered with age. However, in one small study involving single IV doses of 1.5 to 3 mg of lorazepam injection, the mean total body clearance of lorazepam decreased by 20% in elderly subjects 60 to 84 years of age compared to younger adult subjects 19 to 38 years of age.

Administration

For storage information, see the specific product information within the How Supplied section.

Oral Administration

Oral Solid Formulations

Immediate-release Formulations (e.g., tablets)
When given in unequal doses, give the largest dose before bedtime.

Extended-release Oral Capsules (e.g., Loreev XR)
Administer in the morning with or without food.
Do not crush or chew. Have patients swallow the ER capsules whole.
If patient has difficulty swallowing: Contents of the ER capsules may be sprinkled over a tablespoon of cool applesauce and consumed without chewing. Follow with water. Consume all the sprinkled contents within 2 hours. Do not store for future use.

Oral Liquid Formulations

Concentrated Oral Solution (2 mg/mL)
Measure dosage using a calibrated oral syringe/dropper.
Dilute the oral concentrate in water, juice, soda, or semi-solid food (e.g., applesauce, pudding) prior to administration. Administer immediately; do not store for future use.
Storage: Protect from light. Store refrigerated at 36 to 46 degrees F. Discard opened bottle after 90 days.

Extemporaneous Compounding-Oral

Compounded Oral Suspension (1 mg/mL)
Place 180 lorazepam 2 mg tablets in a 12-ounce amber glass bottle. Add the minimum volume of sterile water necessary for tablet dispersion. Shake the bottle until a slurry is formed. Add Ora-Plus and Ora-Sweet to bring the suspension to a concentration of 1 mg/mL (i.e., QS to a total volume of 360 mL). The volume of sterile water required will vary depending on the specific tablets used; this will also result in varying amounts of Ora-Plus and Ora-Sweet depending on the product.
In the chemical stability study, 2 different suspensions were made using the following ingredients:
180 lorazepam 2 mg tablets by Mylan Laboratories, 144 mL of sterile water, Ora-Plus 108 mL, and Ora-Sweet 83 mL.
180 lorazepam 2 mg tablets by Watson Laboratories, 48 mL of sterile water, Ora-Plus 156 mL and Ora-Sweet 146 mL.
Each suspension was divided into 1 oz amber glass bottles for stability testing.
Storage: Suspension is stable for 90 days when refrigerated (4 degrees C) or for 60 days at room temperature (22 degrees C).

Injectable Administration

Do not administer lorazepam injection by intra-arterial injection since arteriospasm can occur which may cause tissue damage and/or gangrene.
Visually inspect parenteral products for particulate matter and discoloration prior to administration whenever solution and container permit.

Intravenous Administration

IV Push
Dilute lorazepam with an equal volume of compatible diluent (0.9% Sodium Chloride Injection, 5% Dextrose Injection or Sterile Water for Injection) immediately prior to use. Mix the contents thoroughly by gently inverting the syringe/vial repeatedly until a homogenous solution is obtained; do not shake vigorously.

For neonatal doses: It may be necessary to make a less concentrated dilution to accurately measure the prescribed dose; some experts recommend dilution to limit the amount of benzyl alcohol administered (some products contain benzyl alcohol 20 mg/mL).
The following dilutions may be prepared using the 2 mg/mL concentration of lorazepam ONLY (do not use lorazepam 4 mg/mL to prepare; precipitation may occur) :
Lorazepam 0.2 mg/mL dilution: Add 1 mL of lorazepam (2 mg/mL) to 9 mL of 5% Dextrose Injection or NS (benzyl alcohol content = 2 mg/mL if using a lorazepam product containing 2% benzyl alcohol).
Lorazepam 0.5 mg/mL dilution: Add 1 mL of lorazepam (2 mg/mL) to 3 mL of 5% Dextrose Injection or NS (benzyl alcohol content = 5 mg/mL if using a lorazepam product containing 2% benzyl alcohol).
After dilution, inject directly into a vein or into the tubing of a freely-flowing compatible IV infusion. Direct IV injection should be made with repeated aspiration to ensure that none of the drug is injected intra-arterially and that perivascular extravasation does not occur.
Inject slowly over 1-5 minutes; do not exceed 2 mg/minute.
Storage: Lorazepam diluted with 5% Dextrose Injection or 0.9% Sodium Chloride Injection at a concentration of 0.2 mg/mL, 0.5 mg/mL, or 1 mg/mL is stable for 24 hours when stored in polypropylene syringes. Dilutions not prepared in a sterile environment should not be stored; discard immediately.

Continuous IV Infusion
NOTE: Lorazepam is not FDA-approved for administration by continuous IV infusion.
Use of glass or polyolefin containers is recommended; polypropylene syringes have also been used. Use of PVC containers results in significant drug loss; PVC administration sets can also be expected to contribute to sorption losses.
Dilute lorazepam injection with a compatible diluent such as 5% Dextrose Injection (preferred) or 0.9% Sodium Chloride Injection to a final concentration of 0.2 mg/mL. For fluid restricted patients, data suggest that a concentration of 0.5 mg/mL or 1 mg/mL is stable for up to 24 hours and may be used.
ASHP Recommended Standard Concentrations for Adult Continuous Infusions: 1 mg/mL.
Lorazepam stability is very specific to the product used and is concentration-dependent. Even at the recommended concentrations, precipitation has occurred in some situations. Carefully evaluate each syringe/bag before administration.
Storage: Lorazepam diluted with 5% Dextrose Injection or 0.9% Sodium Chloride Injection at a concentration of 0.2 mg/mL, 0.5 mg/mL, or 1 mg/mL is stable for 24 hours when stored in polypropylene syringes or glass containers.

Intramuscular Administration

NOTE: For status epilepticus, IV administration is preferred over IM because therapeutic blood concentrations are reached more quickly with IV administration.
When IV access is available, IV is the preferred route of administration due to injection site pain and slower onset associated with IM administration.
When used as a premedication to produce lack of recall, IM lorazepam should be administered at least 2 hours before procedure.
No dilution is needed.
Inject deeply into a large muscle mass (e.g., anterolateral thigh or deltoid [children and adolescents only]).

Maximum Dosage Limits

- **Adults**
  10 mg/day PO; maximum IM and IV dose highly variable dependent upon indication.
- **Geriatric**
  10 mg/day PO; maximum IM and IV dose highly variable depending upon indication.
- **Adolescents**
  Up to 10 mg/day PO for anxiety disorders; 4 mg/day PO for insomnia. Safety and efficacy of extended-release capsules and parenteral lorazepam have not been established. Specific maximum dosage information not available; the dose required is dependent on route of administration, indication, and clinical response.
- **Children**
  12 years: Up to 10 mg/day PO for anxiety disorders; 4 mg/day PO for insomnia. Safety and efficacy of extended-release capsules and parenteral lorazepam have not been established. Specific maximum dosage information not available; the dose required is dependent on route of administration, indication, and clinical response.
  1 to 11 years: Safety and efficacy have not been established. Specific maximum dosage information not available; the dose required is dependent on route of administration, indication, and clinical response.
- **Infants**
  Safety and efficacy have not been established. Specific maximum dosage information not available; the dose required is dependent on route of administration, indication, and clinical response.
- **Neonates**
  Safety and efficacy have not been established. Specific maximum dosage information not available; the dose required is dependent on route of administration, indication, and clinical response.

Dosage Forms

- Ativan 0.5mg Tablet
- Ativan 1mg Tablet
- Ativan 20mg/10ml Solution for Injection
- Ativan 2mg Tablet
- Ativan 2mg/ml Solution for Injection
- Ativan 40mg/10ml Solution for Injection
- Ativan 4mg/ml Solution for Injection
- Lorazepam 0.5mg Oral tablet
- Lorazepam 1mg Oral tablet
- Lorazepam 2mg Oral tablet
- Lorazepam 2mg/1mL Oral solution
- Lorazepam 2mg/1mL Solution for injection
- Lorazepam 4mg/1mL Solution for injection
- Lorazepam Bulk powder
- Loreev XR 1.5mg Extended-Release Capsule
- Loreev XR 1mg Extended-Release Capsule
- Loreev XR 2mg Extended-Release Capsule
- Loreev XR 3mg Extended-Release Capsule

Dosage Adjustment Guidelines

Hepatic Impairment

Lorazepam dosage should be modified based on clinical response and degree of hepatic impairment; a smaller dosage may be sufficient for patients with severe insufficiency. No quantitative recommendations are available.

Renal Impairment

Lorazepam dosage should be modified depending on clinical response and degree of renal impairment. No quantitative recommendations are available. Patients with renal impairment receiving high doses of intravenous lorazepam may be more likely to develop propylene glycol toxicity.

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